

The Signaling Gap:

State-Owned Media and the Strategic Management of Sentiment during the Russian Invasion of Ukraine

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Abstract

Alliance commitments are designed to facilitate cooperation, but they can also prevent states from expressing disagreement through formal diplomatic channels, even when they have reason to dissent. We argue that state-owned media provides an institutionally mediated channel for constrained states to signal distance from a patron's position to foreign governments, occupying a middle ground between formal channels that are too visible and private diplomacy that lacks credibility. We analyze 174,094 news media articles from eleven states before and after the 2022 invasion of Ukraine, classified by a large language model validated against human coders. Using regression discontinuity and difference-in-discontinuities designs, we show that the invasion produced a larger negative shift in sentiment toward Russia in states with Russian security agreements than in neutral states. This occurred despite these states refusing to formally condemn the invasion. Our analysis bridges a gap between political science, which has largely treated state media as a domestic tool, and Western intelligence services, which have monitored it as a window into regime preferences since the Cold War.

Introduction

States that depend on a powerful patron for their security do not always have the luxury of saying what they think. Formal diplomatic channels—UN votes, official statements, public breaks—become liabilities when using them risks retaliation from the very power that guarantees a state’s survival. Yet when a patron’s actions draw international condemnation, silence risks being seen as endorsement. This bind is not unique to international relations. Hirschman (1970) established that when exit is too costly and direct voice risks retaliation, actors seek indirect channels to communicate dissatisfaction. Research on organizational dissent has documented the same logic in hierarchical institutions: when internal voice is foreclosed, subordinates route disagreement to external audiences who have leverage over the principal (Kassing, 1997). In international politics, the problem is sharpest. States bound by security alliances to a patron that has taken internationally condemned actions cannot publicly break with that patron without risking the very security guarantee they depend on. The question is what channels remain available, and when are they credible?

We argue that state-owned media provides such a channel. State media is a formal institution under government control—though how tightly and by what means governments exercise that control varies (Gong et al., 2026)—but it is not directly attributable as a state position in the way a diplomatic communiqué or UN vote is. This makes it usable precisely when directly attributable channels are too costly. Unlike private diplomatic channels, which external audiences cannot verify and can dismiss as cheap talk (Fearon, 1997), state media is publicly observable—giving the signal reach that private communication lacks. The signal is readable because external observers—particularly Western intelligence services—understand state media to reflect regime preferences. This is not a speculative claim. Western governments have monitored state-owned media as a source of intelligence about regime preferences since the Cold War (BBC Monitoring, 2016; Cohen, 1987; Favere, 2024), treating shifts in tone and emphasis as indicators of policy direction within regimes whose formal diplomatic output is tightly

controlled (Williams & Blum, 2018). And the signal is costly: allowing negative coverage of a security patron risks provoking a response, which makes it informative about the sender’s alignment. We show that constrained states exploit this channel to signal distance from a patron’s position to foreign audiences.

Despite this well-established monitoring infrastructure, political science has not formalized the international signaling function of state media. The discipline has produced a substantial literature on how governments use media to manage domestic audiences—setting agendas (McCombs, 2002), shaping attitudes in weak democracies (Enikolopov et al., 2011), and managing information to maintain power (Guriev & Treisman, 2022). But this work is oriented almost entirely toward domestic publics. International relations scholarship, meanwhile, has focused on formal, directly attributable instruments—UN votes, official statements, military deployments—as the channels through which states communicate resolve and commitment (Fearon, 1997; Fuhrmann & Sechser, 2014). The result is a gap: the same outlets that political science studies as tools of domestic control also produce content that intelligence services treat as international signal, yet the discipline has not developed a theoretical framework that accounts for this second function. Our analysis addresses this gap. We demonstrate that the signals practitioners monitor are theoretically structured and empirically identifiable—they follow directly from the logic of alliance constraints, and they discriminate between competing theoretical accounts.

The Russian invasion of Ukraine on February 24, 2022—the largest land war in Europe since World War II—confronted Russia’s formal security partners with a bind between their patron and the West. Formal condemnation risked retaliation from the patron that guarantees their security; silence risked being categorized alongside an internationally condemned aggressor. States bound to Russia through the Collective Security Treaty Organization and other formal security agreements had genuine reasons to oppose the invasion. It violated the sovereignty norms that protect small states and threatened their substantial economic ties with the West. But open condemnation of a security patron that had already demonstrated its willingness to use force in the region

was not a viable option. The invasion thus provides an unusually clear test of our argument: a discontinuity in the information environment, a well-defined alliance structure that constrains formal channels of dissent, and competing theoretical predictions that the data can adjudicate.

We test this argument by exploiting variation in formal security relationships with Russia across twelve states in the region, drawing on a corpus of several hundred thousand newspaper articles published before and after the invasion.¹ Four states were bound to Russia through the Collective Security Treaty Organization or equivalent bilateral security agreements at the time of the invasion. Four occupied broadly neutral positions, maintaining distance from both Russian security structures and full Western integration. The remaining states are NATO members. Articles were translated using neural machine translation and classified with a generative AI model (GPT-4o) to measure sentiment toward Russia at the article level, validating performance against a human-coded gold standard. Our identification strategy uses the discontinuity of February 24 to estimate the effect of the invasion on media sentiment within each group through a regression discontinuity in time design. To compare across groups, we leverage a Regression Difference-in-Discontinuity Design (RDID) (Grembi et al., 2016; Tramontin Shinoki et al., 2024). The discontinuity of the invasion substantially reduces confounding concerns, though comparisons across groups remain correlational in nature. We subject these comparisons to extensive robustness checks, which we detail in the Robustness section.

Our findings build sequentially. We first establish that the invasion produced a significant negative shock to media sentiment toward Russia across the region—without this baseline, the divergence we identify in subsequent tests would be uninterpretable. We then show that this decline was not uniform: independent media responded significantly more negatively than state-owned media, consistent with strategic management

¹The twelve states span three categories of alignment. Russia-aligned: Armenia, Azerbaijan, Belarus, and Kazakhstan. Neutral: Georgia, Moldova, Kosovo, and Serbia. NATO-aligned: Albania, Hungary, and Turkey. Ukraine is included in the dataset but excluded from the main analysis given its direct role as the invaded party.

of coverage by state outlets. The critical test of our argument comes in the comparison across security relationships. A straightforward domestic audience management account would predict that Russia-aligned states suppress negative coverage to protect the alliance relationship. Instead, we find the opposite: state-owned media in states with Russian security agreements shifted more negatively toward Russia than state-owned media in neutral states, despite these same states refusing to condemn the invasion through formal diplomatic channels. This pattern is consistent with constrained states using media to signal distance from their patron precisely because formal channels were unavailable.

This paper makes two primary contributions. First, our analysis demonstrates that the signals practitioners have long monitored are theoretically structured and empirically identifiable. The patterns we document are not idiosyncratic to a single crisis but follow directly from the logic of alliance constraints: states whose formal channels are constrained by dependence on a powerful patron produce systematically different media signals than states whose channels remain open. This suggests that state media sentiment is not merely noise but a legible instrument of international communication, one that the discipline can study with the same rigor it applies to formal diplomatic behavior. Second, we identify an understudied mechanism of institutionally mediated signaling under alliance constraints. The signaling literature has focused on how states communicate resolve to adversaries or commitment to allies through formal channels; we demonstrate that when those channels are foreclosed, states turn to institutionally mediated ones, and that the resulting signals follow predictable patterns rooted in the structure of their security commitments. Methodologically, our approach offers two contributions: the pairing of prompted large language model classification with a regression discontinuity design provides a template for large-scale analysis of media signals, and our matched sensitivity procedure for difference-in-discontinuities—coarsened exact matching combined with the sensitivity framework of Cinelli and Hazlett (2020)—provides a transparent way to assess the robustness of cross-group comparisons when group membership is observational.

The article proceeds as follows. We first develop our theoretical argument and derive three hypotheses about media divergence under alliance constraints. We then present the case, justifying our sample and identification strategy. The following sections describe our data and research design before presenting results. We conclude by discussing implications for how states signal preferences under geopolitical constraint.

Theory and Hypotheses

Formal signaling theory establishes that credible diplomatic signals require the signaling state to incur costs that make bluffing irrational (Fearon, 1997). The logic is straightforward—if a signal is costless to send, it carries no information, because any state could send it regardless of its true intentions. The literature has identified two primary mechanisms through which states generate these costs. The first is audience costs, whereby leaders stake domestic political capital on a public position, making reversal politically expensive (Fearon, 1997). The second involves sinking material costs through actions such as troop deployments, arms transfers, or economic sanctions that are costly to undertake regardless of the outcome (Blankenship & Lin-Greenberg, 2022; Fuhrmann & Sechser, 2014). Both mechanisms share a common feature—they operate through formal, visible, and directly attributable channels. UN votes, official diplomatic statements, military deployments, and public breaks with allies are the instruments this literature has overwhelmingly focused on, precisely because their visibility is what generates the costs that make them credible.

This focus leaves less to say about what happens when those channels are foreclosed. Alliance commitments can create exactly this condition. Alliances do not merely aggregate capabilities against external threats; they also constrain their members (Weitsman, 2004). States that tether themselves to a powerful patron gain security, but they do so by trading autonomy (Morrow, 1991), and one dimension of that lost autonomy is the freedom to publicly dissent. Formal condemnation of a security patron risks what Snyder (1997) identifies as the central dilemma of alliance politics, abandonment. A CSTO member that votes against Russia at the United Nations is not merely casting

a costly vote; it is signaling a willingness to break with the state that underwrites its security, an action that could trigger the withdrawal of that guarantee entirely. The result is that formal channels lock precisely when states have the strongest reason to communicate dissent.

What do constrained states do when formal channels are unavailable? The signaling literature has had less to say about this question, in large part because its theoretical toolkit was built to analyze the instruments where costs are most visible. We argue that constrained states leverage state-owned media as one option for signaling their positions.

The domestic influence of media, including state-owned media, is not understudied. The discipline has produced a substantial literature on how governments use media to manage domestic audiences. News media sets agendas by determining which issues receive public attention (McCombs, 2002) and primes audiences by shaping the criteria they use to evaluate leaders (Iyengar & Kinder, 2010). The framing of an issue shapes how audiences form opinions about it (Chong & Druckman, 2007), and careful framing can have a dramatic impact on public support for government actions, particularly during crises (Gershkoff & Kushner, 2005). Political parties exert significant influence over their followers' opinions and can leverage this to reshape support for policies (Slothuus & Bisgaard, 2021). Work on autocratic media management has extended these insights to non-democracies. Modern autocracies maintain power through sophisticated information control rather than outright censorship (Guriev & Treisman, 2022). The underlying resource wealth of a state shapes the incentives governments face in managing media freedom (Egorov et al., 2009), and state media can powerfully shape attitudes in countries with weak democratic institutions (Enikolopov et al., 2011). Within the post-Soviet space specifically, Oates (2006) argues that state media serves most directly as a tool for elites to engage in agenda setting, highlighting its effectiveness in managing political discourse. This literature has been productive, and much of it applies directly to the states in our sample. But its orientation is almost entirely toward domestic audiences—how governments shape what their own publics think. The possibility that

the same outlets simultaneously communicate to external audiences has received far less attention. Our argument builds on this work. The institutional control this literature documents is precisely what makes state media coverage legible to external audiences as a deliberate choice. The same features that make these outlets effective instruments of domestic agenda-setting also make them usable channels for international signaling.

The practitioners tasked with understanding regime preferences have operated on a different assumption. Western intelligence services began systematically monitoring state-owned media during World War II, with the BBC Monitoring Service established in 1939 and the U.S. Foreign Broadcast Information Service in 1941 (Favere, 2024). During the Cold War, this practice became its own analytical tradition. Western analysts read Soviet and satellite state media not as journalism but as political text, treating shifts in tone, emphasis, and attribution as indicators of factional dynamics and policy direction within regimes whose formal diplomatic output was tightly controlled (Cohen, 1987). Cohen (1987) argues that diplomatic signaling through media is ambiguous by design—it allows governments to communicate positions while preserving deniability, a feature well understood by the intended recipients. This practice has only become more entrenched. Reading open-source material like state media for political signal is now deeply embedded within diplomatic and intelligence institutions (Manor, 2019; Williams & Blum, 2018), and the U.S. Intelligence Community has elevated state media monitoring to a core component of its analytic strategy (USIC, 2024). The British Parliament’s Intelligence and Security Committee described BBC Monitoring as “indispensable” to the national security apparatus (BBC Monitoring, 2016). Nor is this monitoring exclusively Western; Flamer (2023) documents how Hamas systematically monitored Israeli media to extract signals about Israeli policy intentions, suggesting that the practice of reading state-adjacent media as political signal is common knowledge across a range of actors. The institutional infrastructure is not incidental to our argument—it is part of the mechanism. States in our sample operate in an environment where they can reasonably expect that shifts in their state media coverage will be observed, interpreted, and acted upon.

The domestic and international functions of state media are not separate phenomena. The same outlet that manages domestic opinion about a security patron also produces content that foreign analysts monitor for signals of regime preferences. The theoretical tools to understand each function exist in the literature—agenda setting, framing, and autocratic media management on one side; diplomatic signaling, credibility, and monitoring infrastructure on the other—but while the first side has been well-theorized, the latter side has not been assembled into a framework that accounts for this international function. Our argument connects these literatures. We do not claim that all state media operates as an international signaling channel. The argument is sharpest for state media in states where government control over editorial content is well established and widely understood by external observers—conditions that characterize the post-Soviet and nearby states in our sample but that do not describe, for instance, public broadcasters in consolidated democracies. It is precisely because foreign analysts take it as given that outlets like Kazakh or Belarusian state media reflect regime preferences that deviations in coverage become informative.

If state media is a strategic tool, its use as a signaling channel requires that the signal be readable as a deliberate choice and costly enough to be credible. Gehlbach and Sonin (2014) provides a foundation for readability. In their model, governments use state-owned media to advance their agenda, producing systematically higher bias in state outlets than in independent media. This establishes the baseline expectation of alignment from which deviation becomes readable—external observers can distinguish a regime choice from journalistic noise. Readability is what makes state media, rather than independent media, the relevant signaling channel. A regime that simply loosened constraints on independent coverage would produce a shift in tone, but external observers would have no basis for attributing that shift to the regime rather than to editorial independence. State media is readable as a regime choice precisely because external observers understand it to be under government control. States in this context are not merely following elite cues or responding to media pressure on foreign policy (Robinson, 2002); they are deploying a controlled outlet as an instrument

of international communication. There is evidence that state media already operates this way in the region. Peisakhin and Rozenas (2018) show that exposure to Russian state television increased support for pro-Russian candidates in Ukraine’s 2014 elections, demonstrating that state media shapes political attitudes across borders, not only within them.

Readability alone is not sufficient. The signal must also be costly to send. Allowing negative coverage of a security patron risks provoking a response—and a state with no reason to signal distance from the patron would not accept that risk. The costliness of the signal is what makes it informative: it shifts the receiver’s beliefs about the sender’s alignment. The signal’s value to the sender does not require that the receiver learn the sender’s “true” preferences with certainty. It requires only that, after observing the shift in state media coverage, foreign analysts assign a higher probability to the interpretation that the client state is not fully aligned with the patron than they did before. This is the standard signaling logic of Fearon (1997) applied to an understudied channel: the signal is costly (allowing negative coverage risks patron retaliation), therefore informative (a state fully captured by the patron would not accept that risk), and therefore belief-shifting.

The strategic calculus facing constrained states can be stated explicitly. The cost of signaling through state media is the risk of patron retaliation. Where a patron punishes states that break formally with its security framework, even media-level dissent carries risk—the possibility that such dissent could be interpreted as the beginning of a break is the source of cost. The benefit of signaling is shifting other powers’ beliefs about alignment, which may affect sanctions treatment, diplomatic engagement, and the preservation of economic relationships with those powers—relationships that, as we detail in the Case section, are substantial for the states in our sample. The cost of *not* signaling is being categorized alongside the patron by those powers, risking the same diplomatic isolation and economic penalties despite having reasons to oppose the patron’s actions. It is the asymmetry between these costs—the bounded risk of media-level dissent versus the potentially large cost of being lumped in with an internationally

condemned patron—that makes signaling through state media a rational strategy.

The core logic—readability and costliness producing a credible, belief-shifting signal—does not require that the channel be usable repeatedly. Even a single signal is valuable if it shifts the receiver’s beliefs. In practice, however, states face iterated interactions with their patrons, and whether the channel remains available over time depends on additional features of the relationship. Two features support sustainability in this setting. First, state-owned media operates below the threshold of a formal diplomatic break. A patron may punish formal challenges to its security framework, but state-owned media coverage, while observable, does not constitute a formal diplomatic act. The institutional separation between the government and state-owned media outlets provides what Entman (2004) describes as a cascade structure: frames originate with the executive and filter through institutional layers to publication, and these layers—however nominal—provide structural separation. A challenged government can disavow unfavorable coverage as editorial judgment rather than state policy, preserving the patron relationship even as the signal reaches its intended audience. The signal is thus costly enough to be credible but institutionally buffered enough to avoid crossing the threshold that has historically triggered patron punishment.

Second, mutual dependence raises the patron’s cost of responding even to signals it detects. A patron invests substantial diplomatic and military capital in maintaining its alliance framework as a counterweight to other powers, and its client states serve as important economic and strategic partners. Punishing a client state for media-level dissent risks pushing that state toward those powers—precisely the outcome the alliance framework is designed to prevent. The sustainability of the signaling channel thus depends in part on the structure of the patron-client relationship: where both parties are mutually dependent, the patron’s cost of punishing media-level signals is high enough that the channel remains viable.

A natural concern is that observed divergence between state media and official positions reflects internal incoherence rather than strategic communication. Weak democracies and authoritarian regimes are not monolithic. Authoritarians face multiple factions

they must balance (Svolik, 2012), and the elites who surround a ruler are not merely extensions of the leader but actors with their own preferences and survival calculations (Bueno de Mesquita, 2003). Dictators manage these competing interests through institutional bargaining and cooptation (Gandhi, 2008), and state-owned media outlets, as institutions within the regime, may reflect these competing internal voices rather than a single coordinated signal. Expressions of dissent even in the most constrained autocracies are consistent with factional interests rather than top-down direction (Malesky & Schuler, 2010). Our data does not allow us to distinguish between these two mechanisms. But whether state media diverges from formal positions because of strategic choice by leadership or because of elite factional pressure, the observable result is the same: a costly, informative signal of distance from the patron’s position. The receiver’s belief update—the increased probability that the client state is not fully aligned with the patron—is the same regardless of which internal process generated it. The divergence itself is analytically meaningful, and our regression discontinuity design identifies the effect of the invasion on state media sentiment regardless of which internal mechanism produced it. We develop this identification strategy in our methods section.

Our theoretical argument requires that an internationally condemned action by the patron represents a genuine negative shock to media sentiment toward the patron—without this baseline, the divergence we identify in H2 and H3 would be uninterpretable. We first establish this empirically:

Hypothesis 1: An internationally condemned action by the patron produces a significant decline in media sentiment toward the patron across both state-owned and independent outlets.

With this baseline established, we turn to the divergence between state-owned and independent media. State-owned media operates under political constraints that independent media does not face. We therefore expect that state-owned media will manage coverage more carefully than independent media, producing a smaller decline in sentiment toward the patron after the shock. This test is also informative about readability: if state outlets operate under political constraints that shape their coverage, we should

observe a measurably different response to the same shock—and it is precisely this differential that makes state media coverage interpretable as a regime choice rather than independent journalism.

Hypothesis 2: State-owned media will report negative sentiment toward the patron after the shock to a lesser extent than independent media.

A straightforward domestic audience management account would predict that states formally allied to the patron suppress critical coverage of its actions—protecting the alliance relationship means keeping the domestic public from reading negative coverage of the patron. Our signaling argument predicts the opposite: precisely because formal diplomatic condemnation is too costly, these states use state-owned media as an institutionally mediated channel to signal distance to external audiences, producing more negative coverage than neutral states whose formal channels remain open. The comparison to neutral states is critical. States not formally allied to the patron face no alliance constraint on their formal channels and therefore have no signaling need that state media must fill. If anything, neutral states in the patron’s neighborhood have a mild incentive to avoid antagonizing the patron, which should keep state media coverage relatively flat. If state media in patron-aligned states nonetheless shifts more negatively than in neutral states, the pattern cannot be explained by a mechanical response to the shock—it requires the kind of strategic logic our argument describes. That the domestic account and the signaling account generate opposite predictions for H3 provides additional leverage for distinguishing between them.

Hypothesis 3: State-owned media in states formally allied to the patron will respond more negatively to the shock than state-owned media in neutral states.

The Case

The February 2022 invasion of Ukraine was a direct violation of the UN Charter’s prohibition on the use of force against the territorial integrity of a sovereign state (Nations, 1945)—a norm that Russia, as a permanent member of the Security Council, had formally committed to uphold. The international community largely condemned

the attack. The Council of Europe suspended Russian participation, and the UN Security Council took up a draft resolution condemning the invasion, which Russia vetoed (Walker, 2024). Condemnation was the majority response but not a universal one, and the states that withheld it did so for varied economic and geopolitical reasons. The cases at the center of our analysis faced a more specific constraint: bound to the aggressor through formal security agreements, public condemnation meant breaking with the very patron on which their security depended.

The states in our sample fall into three broad categories of alignment, each of which shaped how they could respond. Armenia, Belarus, and Kazakhstan are all members of the Collective Security Treaty Organization (CSTO), Russia’s primary multilateral security alliance in the post-Soviet space. Azerbaijan, while not a CSTO member, signed a Declaration on Allied Interaction with Russia in February 2022, two days prior to the invasion (Huseynov, 2022), placing it in a similar position of formal alignment. CSTO membership creates institutional pressure toward solidarity, making public condemnation of the invasion a direct challenge to the alliance framework itself. At the same time, all four states maintained substantial trade relationships with Western nations, giving them a significant economic stake in opposing the invasion even as their security commitments constrained them from doing so². These competing interests create the dilemma our theory describes. For these states, dependence on Russia forecloses formal condemnation for fear of retaliation, but substantial Western economic and diplomatic ties give these states a real stake in not being seen as complicit in the invasion. Signaling distance through state media offers a way to shift how the West reads their alignment—with potential consequences for sanctions exposure, trade access, and diplomatic standing—while posing a bounded risk of Russian retaliation versus the acute cost of a formal break.

The second category comprises states we treat as broadly neutral: Georgia, Moldova,

²The level of trade reliance expressed as a percent of total trade (imports plus exports) originating from the west (Europe and North America) for each country in 2021 was Belarus: 22%, Armenia: 34%, Kazakhstan: 46%, and Azerbaijan: 72% (Bank, 2021). These trade relationships persisted through the invasion period; Western trade shares for Kazakhstan, Azerbaijan, and Armenia in 2022 were 48%, 73%, and 29% respectively, comparable to 2021 levels.

Kosovo, and Serbia. Each has maintained distance from both Russian security structures and full Western integration, though for different reasons and with different relationships to Russia’s sphere of influence. Georgia and Moldova have both faced Russian territorial incursions in recent years (Cheterian, 2009; Potter, 2022), while Serbia and Kosovo’s distance from both Russia and Western integration reflects different dynamics entirely. Serbia has traditional ties to Moscow, and the tension between those ties and its EU candidacy has shaped its stance on the war (Radeljić & Özşahin, 2023). Kosovo, meanwhile, is dependent on Western recognition for its own sovereignty claim, though it sits outside formal security structures on both sides (Newman & Visoka, 2016). While our neutral states reflect a broad mix of relationships both with the West and with Russia, this mix allows them to function as an appropriate baseline of comparison for states with explicit security agreements. With their formal channels open, neutral states have no signaling need for state media to fill, and a mild incentive to avoid antagonizing Russia should, if anything, keep their coverage relatively flat. This makes them the comparison group for H3, which tests whether aligned-state media shifts more negatively than neutral-state media.

The third category comprises states aligned with Western institutions: Albania and Hungary are both full NATO members, and Ukraine, while not a NATO member at the time of the invasion, had formally declared NATO membership a strategic objective. Turkey presents a more complex case: formally a NATO member since 1952, Ankara declined to join Western sanctions against Russia and instead positioned itself as a mediator between the two parties, reflecting the increasingly independent foreign policy orientation it had pursued since normalizing relations with Moscow after 2016 (Mankoff, 2022). We treat Turkey as NATO-aligned for the purposes of our sample construction, while acknowledging that its behavior during the invasion did not fully mirror that of other alliance members. Because NATO members have an alliance-based reason to oppose the invasion—one that would move their coverage against Russia regardless of any signaling logic—they do not serve as a clean baseline, and the H3 comparison holds them out. Ukraine is included in our dataset but excluded from the main analysis given

its direct role as the invaded party.

This case also offers a way to test our account against an alternative. Because Russia justified the invasion as protection of ethnic Russians abroad, states with large ethnic Russian minorities have their own reason to oppose it—they are the most exposed to the same pretext being turned against them next. If coverage tracks the size of ethnic Russian populations rather than security alignment, this vulnerability, rather than the signaling logic we describe, would explain the pattern. We test this alternative partition in the Robustness section.

The constraints facing Russia-aligned states in February 2022 did not emerge overnight. Russia’s willingness to use force in the region had been established well before the invasion: in 2008 it went to war with Georgia (Allison, 2008), and in 2014 it annexed Crimea following political upheaval in Ukraine and backed separatist forces in the Donbas. The Minsk Agreements, brokered by Western nations, called for a ceasefire and reaffirmation of Ukrainian sovereignty, but violence continued to escalate through the intervening years (Walker, 2023). By November 2021, Russia had mobilized troops along the Ukrainian border, and intelligence from both the U.S. and U.K. was indicating that a full-scale invasion was imminent (Walker, 2023). Russia simultaneously demanded that NATO commit to admitting no new members—explicitly naming Ukraine and Georgia—signaling that states in the region would face consequences for Western alignment (Jazeera, 2022). For CSTO members and Russia-aligned states, this context made formal diplomatic condemnation of the invasion not just politically costly but institutionally constrained.

February 24 nonetheless represents a discrete and dramatic rupture in the information environment. While the troop buildup had proceeded for months, the full-scale invasion—with missile strikes on Kyiv and ground forces crossing the border simultaneously—was of a different order than the slow escalation that preceded it. Russia had conducted large-scale military exercises with Belarus near the Ukrainian border on February 10, with Western intelligence reporting up to 140,000 troops positioned at the border (Walker, 2024). Even so, the decision to launch a comprehensive invasion

rather than a more limited operation represented a threshold that most observers had not fully anticipated.

Although the United States and United Kingdom had taken the unusual step of publicly disclosing intelligence warnings of a full-scale assault in the preceding weeks (Dylan & Maguire, 2022), a large-scale invasion was widely regarded as improbable—even among senior Western officials with access to that intelligence (Banco et al., 2023)—and many governments and analysts, including within the states in our sample, treated a limited operation in the Donbas as the more likely outcome, such that the eventual scale registered as a strategic shock (Götz & Staun, 2022). The media record matches this account. In the state media of Russia-aligned countries—where anticipatory adjustment could be consequential for our design—pre-invasion sentiment trends are small and mixed in sign, and the only detectable discontinuity at the December 7th intelligence warning is a positive increase in sentiment toward Russia, not negative. We discuss this evidence in the Research Design section. This discontinuity is the basis for our identification strategy: we treat February 24 as a break in the media environment and use the variation in coverage before and after it to identify the effect of the invasion on state media sentiment. We describe our data and measurement approach in the following section.

The Data

Our analysis draws on the High-Quality Media from Aid Receiving Countries (HQMARC) corpus, a collection of news articles from prominent domestic media outlets built through human-supervised, source-specific web scraping (Moratz et al., 2025). HQMARC’s design prioritizes comprehensive capture from each source, enabling measurement of both article-level content and source-level patterns of coverage. Articles published in non-English languages were translated using neural machine translation models validated against human translations (Moratz et al., 2025). For the present study, we collected all Russia-related articles published between August 1st, 2021 and August 30th, 2022 from 12 countries and 70 sources across the region. Russia-related

articles were identified using a curated keyword list of over 500 terms associated with Russia, including prominent politicians, government institutions, and companies; the full keyword list is documented in Springman et al. (2026). The 12 countries in our sample span three categories of alignment with Russia, which we describe in detail in the Case section. We omit Ukraine from the analysis given its direct role as the invaded party, leaving 11 countries, 61 sources, and 174,094 articles.

A central distinction in our analysis is between state-owned and independent media. We classify an outlet as state-owned if it is either directly owned and operated by the state or owned and operated by close affiliates of the executive branch, such as family members or cabinet ministers. Ownership was traced through a detailed desk review of media ownership reports, press freedom assessments, and consultations with regional experts, following the procedure described in Moratz et al. (2025). This classification yields 17 state-owned and 44 independent sources across the 11 countries in our sample, with at least one state-owned outlet in each country. The distinction is theoretically central to our argument: external observers—particularly Western intelligence analysts—understand these outlets to be under government control, which is what makes shifts in their coverage readable as regime choices rather than independent editorial judgment (Gehlbach & Sonin, 2014). Figure 1 presents the daily volume of Russia-related articles by media type over the sample period. The sharp spike in coverage at the invasion is visible across both state-owned and independent outlets, though independent media produces substantially more Russia-related content throughout—a difference addressed by the source-level standardization described in the Research Design.

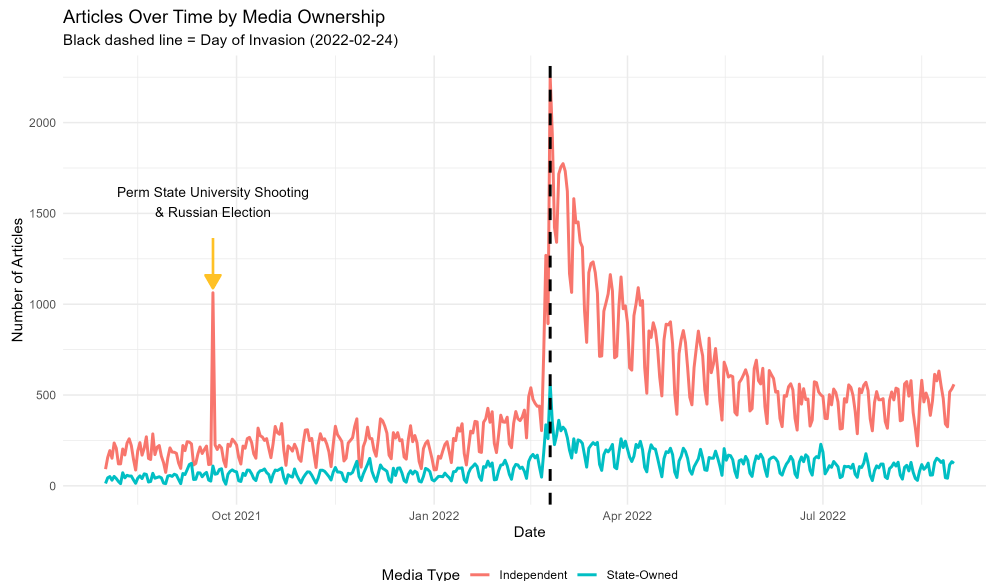


Figure 1: Daily count of Russia-related articles by media ownership type. The black dashed line marks February 24th, 2022.

We coded each article along two dimensions using a zero-shot GPT-4o Turbo classifier applied to the article title and first two sentences: article topic and sentiment toward Russia.³ Our use of a prompted large language model rather than an off-the-shelf sentiment classifier reflects a measurement choice. Standard transformer-based sentiment models (e.g., BERT-based classifiers) capture the overall tone of an article rather than sentiment toward a specific actor. In our setting, this distinction is critical: an article reporting on the invasion could be coded as negative in tone regardless of whether it criticizes Russia, reports neutrally on the conflict, or frames Russia’s actions sympathetically. Our analysis requires sentiment toward Russia specifically—the dimension along which state-owned and independent media are theorized to diverge. The zero-shot GPT-4o classifier allows us to specify this target directly in the prompt, measuring the article’s orientation toward Russia as an actor rather than the valence

³The sentiment and topic classification tasks are substantially simpler than the 20-category civic space classification in Moratz et al. (2025), and zero-shot performance was high enough (F1 of 0.92 for sentiment) that fine-tuning was unnecessary. Applying the classifier to the full article text did not produce significant performance gains, so we used a truncated portion to reduce computational costs. This choice is not merely expedient: Moratz et al. (2025) finds that classification on the title and first two sentences achieves the best performance against human coding on the underlying corpus. Because our gold standard here was likewise coded on this same truncated input, the validation results reflect the classifier’s performance on exactly the text it processes rather than on idealized full-article input.

of the events described. This orientation is a property of the coverage itself—how favorably or unfavorably an article portrays Russia—and not a measure of the outlet’s editorial stance: a state outlet that selects or foregrounds coverage unfavorable to Russia registers as a negative shift whether or not it editorializes in its own voice, which is precisely the channel through which our account expects the signal to operate. Topics were classified as one of five categories: political, economic, military, cultural, or other. Sentiment toward Russia—encompassing the Russian state, military, government ministers, and Vladimir Putin—was coded on a ternary scale: -1 (negative), 0 (neutral), or 1 (positive). We also classify whether Russia is a main actor in each article; this measure is used in supplementary analyses reported in the appendix.

To validate the classifier, we constructed a gold standard of 100 randomly selected articles, adversarially double-coded by the coauthors and debated to consensus. The F1 score for the sentiment classifier is 0.92 and for the topic classifier is 0.80, with confusion matrices reported in Figures A1 and A2. The high F1 scores provide confidence that the gold standard, though modest in size, is adequate for validation. Furthermore, across the validation set, all classifier–human disagreement occurs between adjacent categories—positive/neutral or negative/neutral, with no article coded positive by one coded negative by the other—so classification error compresses sentiment toward neutral rather than reversing it. One feature of the classifier merits note: it is more likely than human coders to classify political articles into the “other” category, an artifact of the overlap between these categories in the data.⁴ Representative examples of articles coded at each sentiment level are presented in the appendix.

Research Design

We leverage two complementary designs to test our hypotheses. First, we use a regression discontinuity design (RDD) to estimate the discontinuous effect of the invasion—

⁴A manual review of 100 randomly sampled articles coded as “other” reveals that the category comprises crime, accidents, and disasters (approximately 20%), COVID-19 and health news (15%), bilateral administrative and logistical matters such as flight routes and travel restrictions (15%), human interest and sports (12%), and miscellaneous international news mentioning Russia tangentially (13%). The remaining approximately 20% are articles with political valence—espionage, journalist persecution, refugee flows—that the classifier codes as “other” rather than “political,” consistent with the overlap noted above.

and the processes it triggered—on media sentiment toward Russia within groups of outlets (Cattaneo & Titiunik, 2022; Cattaneo et al., 2019). Second, we use a regression difference-in-discontinuities (RDID) design to estimate whether the invasion produced *differential* effects on sentiment across groups (Grembi et al., 2016; Tramontin Shinoki et al., 2024). The RDD identifies within-group effects under standard continuity assumptions. Because the running variable in this analysis is time, our design is a regression discontinuity in time. However, this is not an event study—after February 24th there are no untreated outlets to difference against, so within-group identification rests on temporal continuity alone (Hausman & Rapson, 2018). The RDID extends this logic to compare discontinuities across groups—state-owned versus independent media (hypothesis 2) and Russia-aligned versus neutral state media (hypothesis 3)—under a parallel-discontinuities assumption: that, absent the group-specific mechanism, both groups would have experienced the same discontinuity at the cutoff. We attempt to be transparent about this hierarchy of credibility throughout the analysis: within-group discontinuities are causal, while cross-group comparisons leverage the parallel-discontinuities assumption and are subjected to extensive robustness checks, but remain descriptive in nature. The cross-group comparisons the RDID introduces also come with a meaningful cost in statistical power—the comparison underlying hypothesis 3 leaves only 13 state-owned source clusters—which influences the inference procedures we use below.

The Regression Discontinuity Design

We exploit the discontinuity of the Russian invasion on February 24th, 2022 to estimate the effect of the invasion, and everything that resulted, on sentiment toward Russia in news coverage across our sample. In a regression discontinuity in time, the estimand bundles everything that shifts discontinuously at the cutoff: the invasion itself, but also the sanctions, diplomatic ruptures, and changes in coverage composition that the invasion set in motion (Hausman & Rapson, 2018). The forcing variable is the date-time

distance between an article’s publication and the date of invasion.⁵ Because publication times are recorded to the minute and article volume is high, the running variable is dense near the cutoff. This avoids the discrete-running-variable and sparse-data problems that complicate many regression-discontinuity-in-time applications. Treatment is assigned deterministically: all articles published at or after the invasion are treated. Formally, treatment T is assigned by the distance between publication date-time D_p and the invasion D_I :

$$T = \begin{cases} 1 & \text{if } D_p \geq D_I \\ 0 & \text{if } D_p < D_I \end{cases} \quad (1)$$

The key identifying assumption is that potential outcomes are continuous through the cutoff—that is, absent the invasion, sentiment would have evolved smoothly through February 24th. We believe this assumption is reasonable for two reasons. First, the invasion, while anticipated in the weeks prior, constituted a discrete shock: the decision to invade was made by a single actor (the Russian government) and the precise timing was not known *ex ante* to the media outlets in our sample. Consistent with this, state media in Russia-aligned countries show no consistent anticipatory souring: pre-invasion trends are substantively small and mixed in sign, and the only two slopes statistically distinguishable from zero point in opposite directions (Figure 2). A placebo test at the strongest candidate anticipation date also finds no negative discontinuity.⁶ Second, under the stated estimand, same-day events that the invasion itself produced—sanctions announcements, emergency sessions, coverage of the war—are components of the treat-

⁵We discuss our use of publication date-time rather than publication date, and our handling of articles with missing timestamps, in the Forcing Variable section.

⁶We place a placebo cutoff at December 7th, 2021—the date the United States publicly announced that Russia intended to invade. The pooled estimate for aligned state media shows no significant discontinuity (coefficient = 0.108, $p = 0.225$). Only Belarus—the invasion’s staging ground—shows an individually significant discontinuity, and it is *positive* (0.320, $p = 0.025$), suggesting something akin to a rally-around-the-flag response to the announcement that faded as invasion became more plausible (Figure A14). December 7th also lies roughly three times outside every estimated bandwidth (19–28 days), so any response to the warning cannot mechanically contaminate the February 24th estimates. Full results in Appendix Table A13.

ment rather than confounders. The strongest residual threat is a discontinuity from some causally unrelated source affecting sentiment toward Russia across the region’s media on the same date, and we are aware of no plausible candidate.⁷

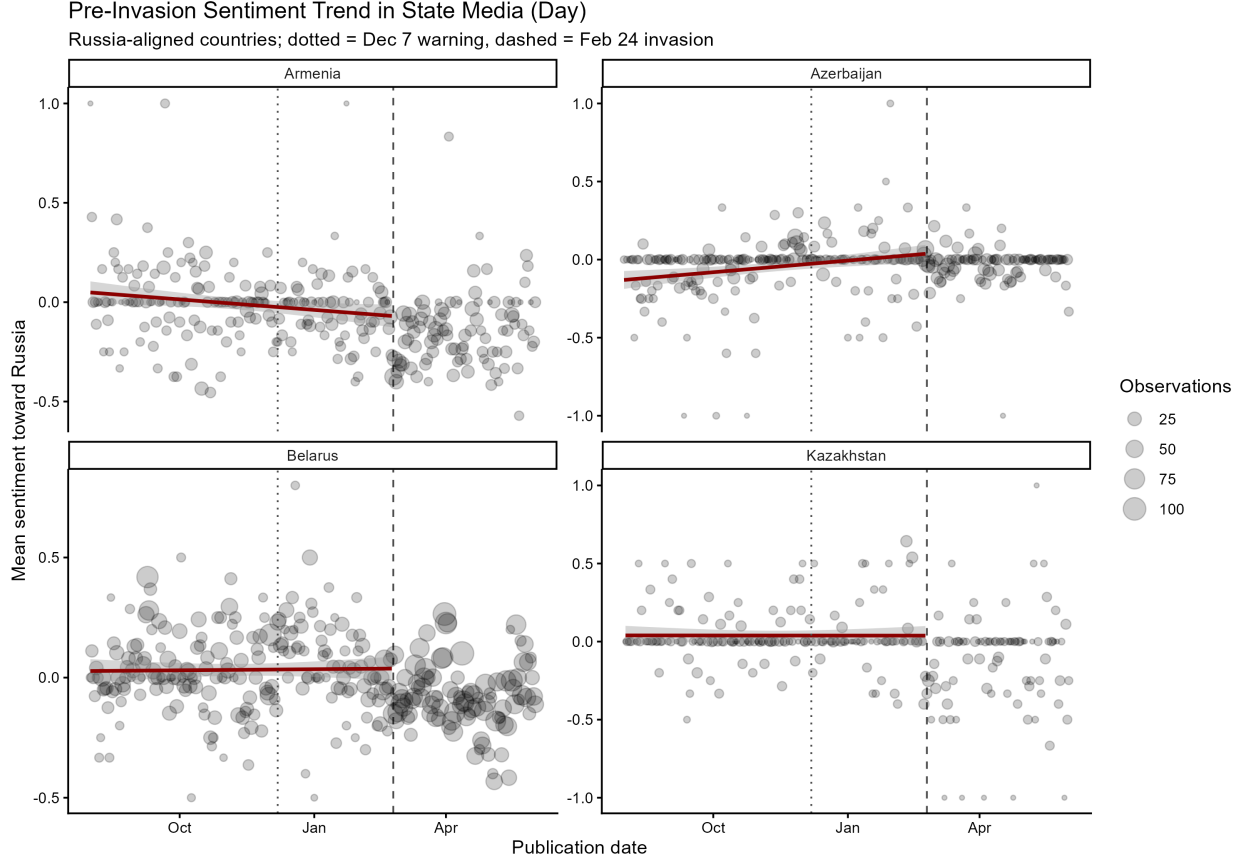


Figure 2: Pre-invasion sentiment trends in state media of Russia-aligned countries. These plots show daily mean sentiment toward Russia by country, with fitted linear trends over the pre-invasion window. While Armenia’s decline and Azerbaijan’s increase are statistically distinguishable from zero, overall these trends are substantively small and mixed in sign. Points are sized by daily article count, with the dotted line marking December 7th, 2021, and the dashed line marking February 24th, 2022.

Under continuity, we can identify the local average treatment effect (LATE) of the invasion on sentiment at the cutoff:

⁷Our estimates should be, if anything, conservative with respect to immediate pre-invasion escalation. Russia’s recognition of the Donetsk and Luhansk republics on February 21st and the first Western sanctions on February 22nd fall on the left side of the cutoff within the typical bandwidths used in our analysis, so any sentiment response to them is absorbed into the pre-invasion baseline, reducing the measured jump.

$$\widehat{\gamma_{\text{rdd}}} = \lim_{D_p \downarrow D_I} E[Y_p(1) \mid D_p = D_I] - \lim_{D_p \uparrow D_I} E[Y_p(0) \mid D_p = D_I] \quad (2)$$

where $Y_p(1)$ and $Y_p(0)$ represent article-level sentiment toward Russia under the post-invasion and pre-invasion potential outcomes, respectively. For hypothesis 1, this estimand is identified using `rdrobust` (Cattaneo et al., 2019) with the full set of fixed effects (country, source, and topic) and MSE-optimal bandwidth selection (Calonico et al., 2020). We use HC1 heteroskedasticity-robust standard errors clustered at the source level for all RDD specifications.

The Regression Difference-in-Discontinuities Design

Hypotheses 2 and 3 ask whether the invasion produced differential effects across groups. The RDID extends the RDD by comparing the estimated discontinuity in one group to the estimated discontinuity in another (Grembi et al., 2016; Tramontin Shinoki et al., 2024). If state-owned media responded differently from independent media (H2), or if Russia-aligned state media responded differently from neutral state media (H3), the RDID will detect this as a significant difference in the magnitude of the two discontinuities.

No nonparametric framework for the RDID currently exists. We therefore estimate the RDID using OLS within the MSE-optimal bandwidth, applied to a fully interacted specification:

$$\begin{aligned} Y_p^z = & \beta_0 + \beta_1 D_p + \beta_2 T_p + \beta_3 D_p T_p \\ & + \beta_4 G_p + \beta_5 T_p G_p + \beta_6 D_p G_p + \beta_7 D_p T_p G_p + \gamma_{\text{topic}} + \varepsilon_p \end{aligned} \quad (3)$$

where Y_p^z is the z-scored sentiment of article p (standardized within source; see the Standardization section), D_p is the running variable, T_p is the treatment indicator, G_p is the group indicator (state-owned vs. independent for H2; Russia-aligned vs. neutral

for H3), and γ_{topic} is a vector of topic fixed effects. The error term ε_p is clustered at the source level. The RDID estimate is β_5 : the differential shift in sentiment at the cutoff for the group of interest relative to the comparison group. We detail the rationale for the z-scored outcome, the bandwidth selection procedure, and inference with few clusters in subsequent sections.

The identifying assumption for the RDID is structurally analogous to the parallel trends assumption in difference-in-differences, applied to discontinuities rather than levels. The specification absorbs level differences between groups (β_4) and differential trends in the running variable (β_6). What remains is the assumption that, absent the group-specific mechanism we theorize, both groups would have experienced the same discontinuity at the cutoff. This is the parallel-discontinuities assumption. Unlike parallel trends, this assumption places no restriction on how the groups trend toward the cutoff. Pre-existing anticipatory drift, even if that drift differs substantively across groups, is absorbed by the trend terms and cannot produce a spurious differential jump (Tramontin Shinoki et al., 2024). Indeed, if anything, anticipation works against our hypotheses—any pre-invasion adjustment by aligned state media moves their sentiment toward the neutral group before the cutoff, shrinking the likelihood of a differential discontinuity that could remain to be measured. The threat to identification is a factor that produces a differential *jump* in sentiment at the invasion date, is correlated with group membership, but operates through a channel other than the one our theory describes. We discuss the conditions under which RDID estimates admit a causal interpretation—and why our specification is best understood as a strongly informative comparison rather than a causal claim about group membership—in the Interpreting the RDID section. We detail our sensitivity analysis and robustness checks, which directly address the plausibility of confounding threats, in the Robustness section.

Identification and the Role of Covariates

Isolating the effect of the invasion on sentiment requires distinguishing the pathways through which the invasion can affect our outcome. The invasion does not only change

how outlets write about Russia—it also changes *what* they write about. If military topics spike after the invasion and military coverage is inherently more negative toward Russia, then the estimated discontinuity will conflate genuine shifts in editorial tone with a mechanical recomposition of the article pool. This is a compositional threat to identification, not a confounding variable in the traditional sense: the invasion causes both the topic shift and the sentiment shift, but our quantity of interest is the direct effect on sentiment, not the indirect effect operating through topic composition. Figure A7 in the Appendix presents this structure as a directed acyclic graph.

A naive solution—simply controlling for topic—is insufficient. Topic is a collider on the path between source characteristics (country, ownership) and sentiment. Conditioning on topic alone therefore opens a backdoor path through country and ownership, which can introduce bias. The solution requires blocking all three pathways simultaneously: topic, country, and ownership.

Within our sample, each outlet has a fixed country and ownership type. Source identity therefore subsumes both country and ownership as a finer partition. Controlling for source blocks the backdoor paths through country and ownership while also absorbing source-level differences in topic profiles. Figure 3 illustrates this solution. Topic fixed effects in the regression then absorb remaining within-source variation in topic composition at the cutoff.

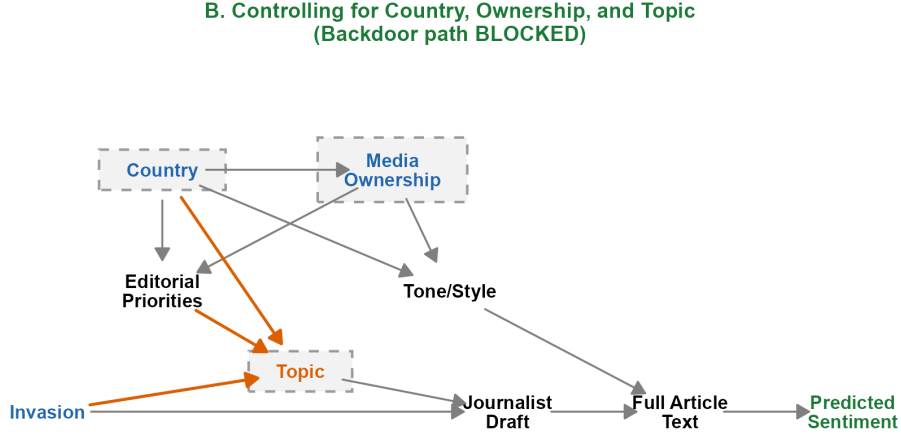


Figure 3: Directed acyclic graph illustrating the identification solution. Controlling for source blocks the backdoor paths through country and ownership. Topic fixed effects absorb compositional shifts within sources.

However, covariate adjustment in the RDD framework is only a valid identification strategy if the covariates being adjusted for do not themselves exhibit a sharp discontinuity at the cutoff (Cattaneo et al., 2023). If they do, the fixed effects absorb part of the treatment effect rather than isolating it. Here, this concern is substantive: the invasion plausibly produces a sharp increase in the share of articles coded as military in nature. If military articles spike discontinuously at February 24th, topic fixed effects may partial out a component of the genuine treatment effect. Relatedly, article volume itself increases sharply after the invasion, changing the composition of the sample at the cutoff. Since outlets do not choose their publication dates in response to the invasion threshold per se, manipulation of the forcing variable is not a concern in this setting. Therefore, this volume change reflects a compositional shift rather than a threat to the assignment mechanism—selection into publication is itself part of the treatment—and

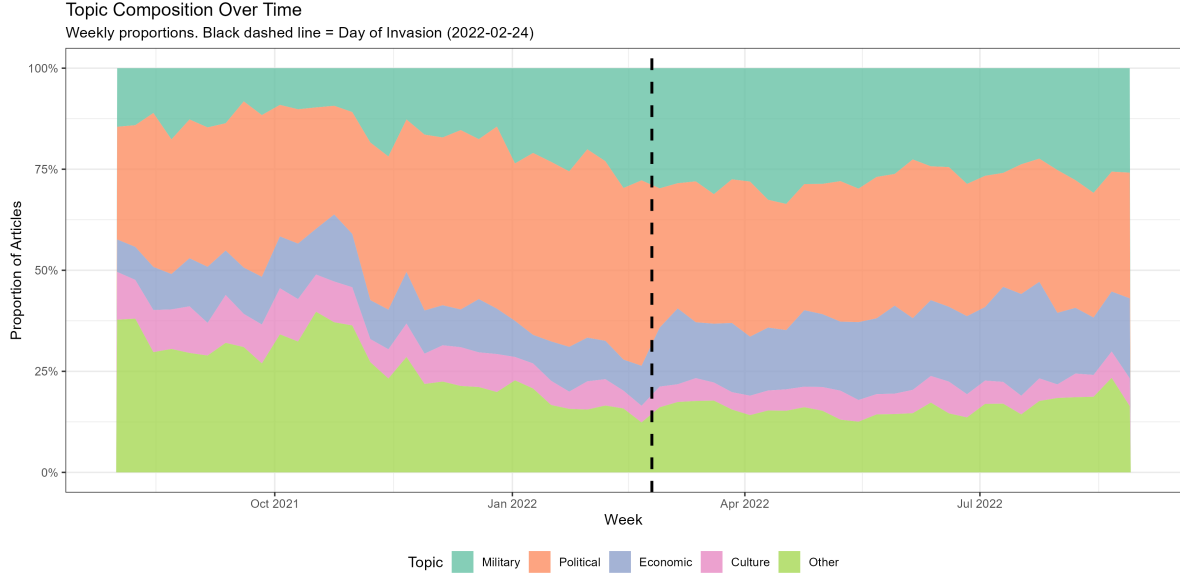


Figure 4: This figure shows the proportion of articles each topic comprises over time, measured weekly. Political topics dominate the pre-invasion period, but leading up to the invasion, military articles begin to grow in prominence. Other articles experience a steady decline before the invasion, but are flat afterwards. After the invasion, there is an increase in economic articles that draws from political articles.

is addressed by the topic-level checks described below.

We address the compositional concern in two ways. First, we include specifications in which articles coded as military are omitted entirely, removing the compositional channel by construction. Second, we estimate the RDD within each topic separately, allowing us to observe how the discontinuity operates *within* topic categories rather than across them. If the discontinuity is present within non-military topics, the compositional threat cannot account for the main result. Both checks are reported in the Robustness section. Figure 4 shows how the composition of articles changes over time. Military coverage grew steadily in the months preceding the invasion as the troop buildup progressed, and while its share increases further after February 24th, the shift is consistent with the pre-existing trend. The more pronounced compositional change is a reallocation from political to economic coverage in the weeks immediately following the invasion.

Our use of topic fixed effects also raises the question of what the topic-adjusted estimates identify. Because topic is measured post-treatment, these regressions do not

recover the total effect of the invasion (Montgomery et al., 2018). Instead, they recover a controlled direct effect—the sentiment shift with topic composition held fixed—which requires a sequential-unconfoundedness assumption beyond continuity (Acharya et al., 2016). We discuss this assumption in the Extended Research Design appendix and explore the impact of using topic fixed effects in the Robustness section.

A downstream consequence of source-level control is that source identity is collinear with our key partitions of interest—whether an outlet is state-owned or independent (H2) and whether it belongs to a Russia-aligned or neutral state (H3). These group memberships are fixed within source, which mechanically prevents direct estimation of the RDID interaction when source fixed effects are included in the regression. We address this constraint in the following section.

Standardization and the Z-Score Specification

The identification strategy developed in the preceding section establishes that source-level control is necessary to block the backdoor paths through country and ownership. However, including source fixed effects in the RDID regression is infeasible: because each source belongs to exactly one group (state-owned or independent, Russia-aligned or neutral), source fixed effects perfectly absorb the group indicator G_p and mechanically prevent estimation of the RDID interaction β_5 .

We resolve this constraint by standardizing the dependent variable at the source level prior to estimation. For each source s , we compute:

$$Y_p^z = \frac{Y_p - \bar{Y}_s}{\sigma_s} \quad (4)$$

where $\bar{Y}_s$ and σ_s are the mean and standard deviation of sentiment for source s , computed over the full sample period. The demeaning component $(Y_p - \bar{Y}_s)$ is mathematically equivalent to a source fixed effect by the Frisch-Waugh-Lovell theorem (Lovell, 1963): it removes all source-level intercept differences, including those attributable to country, ownership, and baseline editorial orientation. Critically, because the group

indicator is absorbed by demeaning rather than by a fixed effect in the regression, G_p remains estimable in the RDID specification, and the interaction β_5 can be identified.

The division by σ_s addresses a second concern. Although all articles in our sample are coded on a ternary sentiment scale $(-1, 0, 1)$, sources differ substantially in baseline variance—some produce predominantly neutral coverage while others are more polarized. In a linear specification, high-variance sources receive disproportionate influence on the estimated coefficients (Aronow & Samii, 2016). Without adjustment, the RDID estimate would be biased toward the experience of whichever sources happen to exhibit the greatest variation in sentiment, rather than reflecting the average response across sources. Dividing by σ_s ensures that each source contributes proportionally to the estimate, regardless of its baseline variance. It also aids interpretation: coefficients from the z-scored specification represent shifts in within-source standard deviations, providing a natural scale for comparing effect magnitudes across groups with different baseline distributions.

Two features of this standardization merit emphasis. First, the z-score is computed over the *full sample period*, not within the estimation bandwidth. This is a deliberate choice: if the standardization were computed within the bandwidth, the mean and variance would themselves be functions of the bandwidth, complicating interpretation and introducing a dependency between bandwidth selection and the dependent variable. Computing the z-score over the full sample period ensures that the standardization is a fixed transformation of the data, independent of the estimation procedure.

Second, we validate the z-score specification against the full fixed-effects approach in the one setting where both are estimable. For hypothesis 1—the within-group RDD—we first estimate the discontinuity using `rdrobust` with the complete set of fixed effects (country, source, and topic). We then re-estimate hypothesis 1 using the z-scored outcome with topic fixed effects only. The two approaches produce directionally equivalent results, indicating that the z-score specification adequately absorbs source-level heterogeneity. Having established this equivalence in the setting where both approaches are available, we proceed with the z-score specification for hypotheses 2 and 3, where the

full fixed-effects structure cannot accommodate the group interaction.

The Forcing Variable

The forcing variable in our design is the distance between an article’s publication date-time and the *date* of invasion (February 24th, 2022). We measure distance from the date rather than a specific moment of invasion to maintain a sharp design: all articles published on or after February 24th are treated, with the within-day publication time providing variation in proximity to the cutoff.⁸ A standard approach in time-based RDD would use calendar date alone as the running variable. However, calendar date creates severe probability mass points: hundreds of articles may share the same publication date, producing substantial heaping that complicates both estimation and inference (Cattaneo et al., 2024). Incorporating publication time effectively eliminates mass points—exact timestamp ties are rare, rendering the forcing variable continuous or very nearly so.

Not all articles in our corpus have observed publication times. For these articles, we impute a publication time using kernel density estimation over the distribution of observed publication times for the same source. If a source has too few observed timestamps to estimate a reliable density, we instead use the country-level distribution of publication times. This approach preserves the publishing rhythm of each source as a prior: an article from a source that typically publishes in the morning will receive a morning timestamp, not a time drawn from a uniform distribution. Each article receives a single imputed time, drawn once and held fixed across all analyses.

We subject this approach to two robustness checks, reported in the appendix. First, we replicate all analyses using publication date rather than date-time. Mass points are present in this specification, but results are substantively identical to the date-time results, suggesting that heaping affects precision but does not introduce bias. Second, we replicate all analyses after dropping articles without observed publication times entirely. Results again hold, indicating that the imputation procedure is not driving

⁸Using a specific moment (e.g., the hour the invasion began) would require treating articles published on February 24th before that moment as untreated, introducing fuzziness in treatment assignment.

our findings.

Estimation and Inference

Our estimation procedures differ across the two designs. For hypothesis 1, we estimate the within-group RDD using `rdrobust` (Cattaneo et al., 2019), which implements local polynomial regression with MSE-optimal bandwidth selection (Calonico et al., 2020), bias-corrected confidence intervals, and cluster-robust standard errors at the source level.

For hypotheses 2 and 3, no nonparametric framework for the RDID currently exists, and the fully interacted specification with topic fixed effects exceeds what `rdrobust` can accommodate. We therefore estimate the RDID using OLS, which requires selecting a bandwidth separately. Because `rdbwselect` also cannot handle the fixed-effects structure directly, we first residualize the z-scored outcome on topic fixed effects and the group indicator using `feols`, then pass these residuals to `rdbwselect` with a triangular kernel, source-level clustering, a local linear polynomial ($p = 1$), and the MSE-optimal selection rule. This ensures the bandwidth is selected on the variation that the RDID specification actually exploits. Having obtained the optimal bandwidth, we filter the data to observations within this window and estimate the RDID via `feols` using the specification in Equation 3. Results are robust across bandwidths ranging from half to approximately 1.5 times the optimal bandwidth (see appendix).

Inference with Few Clusters

A central challenge in our design is that the number of clusters—sources—is small. For the full-sample RDID (hypothesis 2), we have approximately 60 source-level clusters. For the state-media-only RDID (hypothesis 3), we have 13.⁹ Standard cluster-robust variance estimators assume that the number of clusters is large; with few clusters, they can substantially over- or under-reject (Cameron & Miller, 2015).

⁹While sources nest within countries, and country-wide shocks could induce dependence across sources that source-level clustering ignores, we choose to cluster at the source level for two reasons. First, the source-level clustering matches the level of the standardization we apply throughout the analysis. Second, clustering at the source level preserves more clusters for our bootstrap. However, we discuss alternative specifications of our results using country-level clustering in the Robustness section.

We address this using the wild cluster restricted (WCR) bootstrap of Cameron et al. (2008) with Webb six-point weights and 4,999 replications, imposing the null hypothesis. These are preferred to the default Rademacher weights when the number of clusters is small (MacKinnon & Webb, 2017), as in our hypothesis 3 sample ($G = 13$). The WCR bootstrap provides asymptotic refinement relative to cluster-robust standard errors when the number of clusters is small, and the restricted version (imposing the null) has better finite-sample properties than the unrestricted alternative (MacKinnon & Webb, 2017). We report the bootstrap p -values for all RDID specifications. For the `rdrobust` models (hypothesis 1), inference relies on the package’s internal cluster-robust standard errors at the source level.

As a further check on inference, we estimate fully saturated source-by-topic RDDs in the appendix—a separate regression for each source-topic cell—and aggregate using both a pairs bootstrap at the source level and a Bayesian bootstrap with Dirichlet-distributed weights (Oganisian & Roy, 2021; Rubin, 1981). Because each cell constitutes its own regression, this approach sidesteps the clustering problem entirely, though at the cost of reduced power. Results are consistent with the main specifications.

Finally, we note a consideration raised by Abadie et al. (2022): if the sources in our sample constitute the population of relevant outlets in these countries rather than a sample drawn from a larger superpopulation, then clustering may be overly conservative. Our sample is close to this setting—we collect all major outlets, not a random draw from a larger universe of potential sources—but we do not rely on this argument and report clustered inference throughout.

Interpreting the RDID

The within-group RDD estimates for hypothesis 1 are causal under the standard continuity assumption: absent the invasion, sentiment would have evolved smoothly through the cutoff. We have discussed the grounds for this assumption in the Regression Discontinuity Design section. The interpretive question that requires further discussion is what the RDID estimates for hypotheses 2 and 3 identify.

The RDID compares the magnitude of the discontinuity in one group to the magnitude of the discontinuity in another. As Bansak and Nowacki (2022) emphasize in the context of RDD heterogeneity, it is important to distinguish between estimates that are *conditional on* group membership and estimates that are *causally driven by* group membership. The former—the empirical fact that Russia-aligned state media responded differently from neutral state media—is identified under the parallel-discontinuities assumption described in the RDID section. The latter—the claim that Russia-alignment itself caused the differential response—requires stronger assumptions.

Tramontin Shinoki et al. (2024) develop the formal conditions under which the RDID supports causal identification, building on Grembi et al. (2016). Causal identification requires that the comparison be expressible as a *discontinuity of differences*—the same units observed in two periods, so that differencing within units removes any pre-existing discontinuity before the RDD is estimated (as in Grembi et al.’s municipalities observed before and after a fiscal reform). Toral (2024) applies the same logic to bureaucrats within schools. Our setting does not have this panel structure. We observe different groups at a single discontinuity—a *difference of discontinuities*. Without within-unit differencing to absorb pre-existing group differences at the cutoff, the RDID cannot rule out that the differential response reflects a factor correlated with group membership rather than the mechanism our theory describes.

We therefore interpret our RDID estimates as conditional on group membership—the differential response is a robust empirical pattern, not a causal effect of Russia-alignment per se. Fu and Slough (2026) formalize when a differential effect like ours supports a claim about the mechanism—namely, the characteristic driving the moderation must not also moderate other channels of response. In our case, alignment must shape the response to the invasion through the signaling channel alone. Three features of our analysis provide evidence that the pattern is driven by the mechanism our theory describes rather than by an omitted confounder correlated with group membership. First, a sensitivity analysis using the framework of Cinelli and Hazlett (2020), applied after coarsened exact matching on publication date and topic, yields a robust-

ness value of 1.9%. As we detail in the Robustness section, nullifying the differential would require a confounder more strongly associated with sentiment than the invasion itself. Second, our theoretical framework generates opposing predictions for the signaling and domestic audience management accounts, and the data are consistent with the signaling account and inconsistent with the domestic account. An omitted confounder would need to not only produce the observed differential but also do so in the specific direction that contradicts our theory. Channels that would produce the differential in the same direction—for example, differential economic ties to Russia causing more direct exposure to economic fallout—cannot be excluded by sign alone. However, the sensitivity analysis bounds how strong any such channel would need to be. Channels operating through anticipation—such as differential foreknowledge of the invasion—cannot produce a differential jump at the cutoff and, as discussed above, are absorbed by the group-specific trend terms. Third, we test an alternative group partition—splitting states by the size of their ethnic Russian population rather than by Russia-alignment—and find no differential response (Table 6). If the RDID result were driven by a confounder correlated with proximity to Russia or cultural ties rather than by the alliance constraint our theory identifies, we would expect the ethnic Russian partition to produce a similar pattern. It does not.

A final consideration concerns measurement. Fu and Slough (2026) caution that non-affine transformations of an outcome—including coarse categorical coding like our ternary sentiment scale¹⁰—can generate differential responses mechanically when groups sit at different distances from the scale’s bounds. Two features of our data weigh against this concern. First, neutral-state media sentiment begins far from the scale floor and barely moves after the invasion, while independent media in the same countries drop sharply (Appendix Figure A8), so the absence of a neutral-state response is not reflective of a floor effect. Second, classifier disagreement with human coders occurs only

¹⁰We intentionally retain only the categorical classifications of sentiment. Verbalized confidence and post-reinforcement learning from human feedback token probabilities are both poorly calibrated (OpenAI et al., 2023; Xiong et al., 2023), so we do not attempt to construct a finer-grained outcome from unreliable confidence measures.

between adjacent categories (positive/neutral and negative/neutral), never across the range (positive/negative), so measurement error should only attenuate the observed differentials rather than manufacturing them (see the Data section).

Analysis

We present results sequentially, beginning with the baseline effect of the invasion on media sentiment across the region (H1), then examining divergence between state-owned and independent media (H2), and finally testing whether the pattern of divergence varies with formal security alignment (H3).

H1: The Invasion as a Negative Shock to Media Sentiment

Table 1 presents our RDD estimates of the effect of the invasion on sentiment toward Russia across all media in the sample. Model 1, the simplest specification without covariates, estimates a decline in raw sentiment of 0.130 ($p < 0.001$). Model 2 adds topic fixed effects; Model 3 further adds source fixed effects; Model 4 includes topic, source, and state fixed effects. Across these specifications, estimates range from 0.106 to 0.115, all significant at $p < 0.001$. The stability of the estimate across these specifications indicates that the result is not driven by differences across sources, countries, or topic composition.

Models 5 through 7 use the z-scored outcome, which centers sentiment at the source level and scales by within-source standard deviation, providing a standardized measure of the shift. Model 5 estimates a decline of 0.217 within-source standard deviations ($p < 0.001$). Model 6 omits articles classified as military in topic, testing whether the result is an artifact of the mechanical influx of negatively-valenced war coverage after February 24th. The estimate attenuates slightly to 0.184 standard deviations but remains highly significant, consistent with the sentiment shift extending beyond military reporting to political, economic, cultural, and other coverage. Model 7 drops NATO-allied outlets from the sample; the result is robust to their exclusion.

Figure 5 presents the discontinuity visually for the non-military sample, correspond-

ing to the Model 6 specification. The break at February 24th is clearly visible, with mean daily sentiment dropping from approximately -0.15 pre-invasion to approximately -0.30 immediately after. The post-invasion trend slopes upward, consistent with a gradual recovery in sentiment over the following months, but the initial discontinuity is unambiguous. Additional robustness checks, including bandwidth sensitivity and alternative forcing variable constructions, are reported in Table A1.

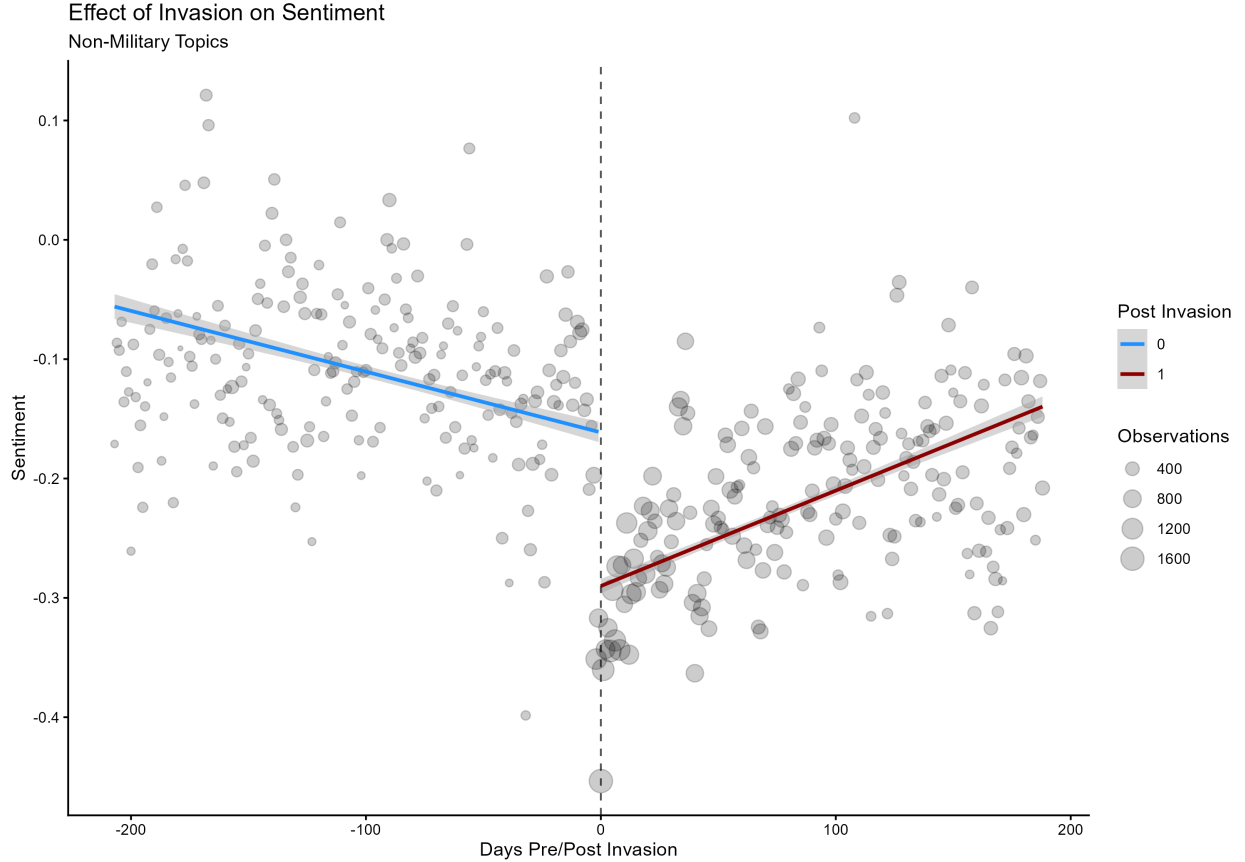


Figure 5: RDD estimate of the effect of the invasion on sentiment toward Russia, non-military articles. Each point represents one day, sized by the number of articles published. The dashed line marks February 24th, 2022.

H2: State-Owned Media Manages Coverage More Carefully

Having established that the invasion produced a sharp negative shift in sentiment across the region, we now test whether state-owned and independent media responded differently. Hypothesis 2 predicts that state-owned media, operating under political constraints that independent outlets do not face, will produce a smaller decline in sentiment

Table 1: H1 RDD — Effect of Russian Invasion on Media Sentiment (Probabilistic Time)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Invasion	-0.130*** (0.022)	-0.115*** (0.021)	-0.106*** (0.015)	-0.106*** (0.015)	-0.217*** (0.036)	-0.184*** (0.033)	-0.284*** (0.038)
Topic FEs	-	X	X	X	X	X	X
State FEs	-	-	-	X	-	-	-
Source FEs	-	-	X	X	-	-	-
Source Z-Score	-	-	-	-	X	X	X
Clusters	61	61	61	61	61	61	45

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

All standard errors are clustered at the source level. Models 1-4 use the optimal bandwidth. Models 5-7 use a Z-Score adjustment instead of fixed effects. Model 6 omits Military Topics. Model 7 omits NATO allies.

toward Russia.

Table 2 presents results in two stages. Models 1 and 2 report separate RDD estimates for state-owned and independent media, restricted to non-military articles. Independent media exhibits a decline of 0.214 within-source standard deviations ($p < 0.001$), while state-owned media declines by 0.109 standard deviations ($p < 0.1$). Both shift negatively, but independent media responds roughly twice as strongly as state-owned media.

Models 3 through 6 estimate the RDID interaction directly. The state-owned $\times$ treatment coefficient captures the extent to which state media’s response to the invasion differs from that of independent media. Model 3, using the full sample at the optimal bandwidth, estimates an interaction of 0.080—positive, indicating that state media was less negative, but imprecise. Model 4 omits military articles, sharpening the estimate to 0.14 ($p < 0.1$). Models 5 and 6 restrict the sample to non-NATO countries, removing outlets whose institutional alignment with the West provides a distinct basis for opposing the invasion that is unrelated to the state-versus-independent dynamic our theory describes. In this sample, the interaction reaches 0.222 ($p < 0.05$), and the result is unchanged when coarsened exact matching on publication date and topic is applied (Model 6). The matched specification indicates that the divergence is not driven by observable differences in the timing or subject matter of coverage across state-owned

and independent outlets.

Figure 6 presents the discontinuity by media type for the non-military sample. The visual pattern is consistent with the regression estimates: both state-owned and independent media shift downward at the cutoff, but independent media drops more sharply and recovers more slowly. The gap between the two post-invasion trend lines reflects the RDID interaction.

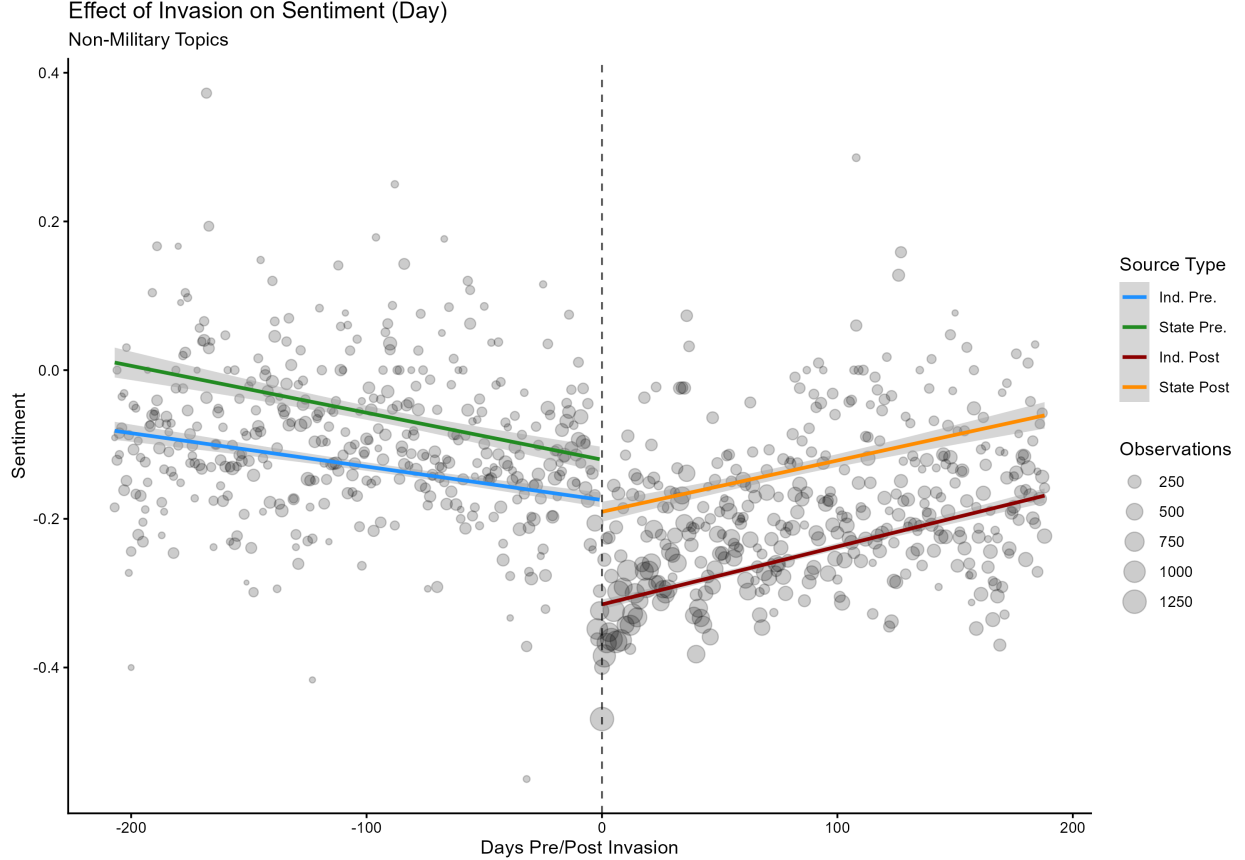


Figure 6: RDD estimates of the effect of the invasion on sentiment toward Russia by media type, non-military articles. Each point represents one day, sized by the number of articles published. The dashed line marks February 24th, 2022.

A single pair of articles illustrates the divergence Hypothesis 2 estimates and offers a spot-check on the classifier. In the days around the invasion, Serbia’s state broadcaster and an independent outlet covered the same war from the same country, but their coverage of Russia pulled in opposite directions. On February 25, the independent outlet N1 ran “Canak: Putin started the attack on the same day that Hitler presented his

Table 2: H2 Diff-in-Disc — State Ownership Interaction, Z-Score (Probabilistic Time)

	(1) State-Owned‡	(2) Ind.‡	(3)†	(4)†	(5)†	(6)†
Invasion	-0.1090+	-0.214***	-0.253***	-0.240***	-0.347***	-0.348***
	(0.0583)	(0.034)	(0.036)	(0.031)	(0.027)	(0.026)
State-Owned x Treat			0.080	0.14+	0.222*	0.222*
			(0.063)	(0.06)	(0.069)	(0.069)
Source Z-Score	X	X	X	X	X	X
Clusters	17	44	60	60	43	43
Num.Obs.	30978	99510	40710	29278	31405	31379
R2			0.045	0.028	0.024	0.024
R2 Adj.			0.045	0.028	0.023	0.023

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

All models use a source-level Z-Score normalization. Models 1-2 show separate RDD estimates for state-owned and independent media, filtered to omit military topics. Models 3-6 estimate the state-owned x treatment interaction. Model 3 uses the optimal bandwidth on the full sample. Model 4 omits military topics. Model 5 omits military topics and NATO allies. Model 6 applies exact matching on topic and publication date to Model 5’s sample. † Standard errors computed via wild cluster restricted (WCR) bootstrap (Cameron, Gelbach & Miller 2008) with Webb six-point weights (B=99999, null imposed), clustered by source domain. ‡ Cluster-robust SEs from rdrobust.

program,” reporting a Serbian politician’s charge that Putin had launched the invasion on the anniversary of Hitler’s Nazi program—coverage the classifier coded negative.¹¹ A day later, the state broadcaster RTS ran “Bocan-Harchenko: Russia understands Serbia’s position, does not require anything from Belgrade,” a report built around the Russian ambassador’s assurances to Belgrade—coverage coded neutral. Both articles concern the same war, yet the classifier separates the one oriented negatively toward Russia from the one that is not. The medium is held constant—both are Serbian outlets covering the invasion in real time—so the gap in orientation tracks ownership rather than the event or the country’s underlying disposition. This pair shows the state-versus-independent divergence in action.

Table 3 disaggregates the RDID by topic category. The state-owned $\times$ treatment interaction is positive across all five topics, indicating that the pattern of muted state media response is not confined to a single subject area. The interaction is statistically

¹¹Translations are machine-generated and reproduce the title and opening sentence supplied to the classifier; the source texts are in Serbian.

Table 3: H2 Diff-in-Disc by Topic — Z-Score (Probabilistic Time)

	Political	Military	Economic	Cultural	Other
Invasion	-0.213*** (0.027)	-0.422*** (0.068)	-0.423*** (0.066)	-0.547*** (0.071)	-0.269*** (0.039)
State Owned x Treatment	0.155 (0.079)	0.012 (0.102)	0.29* (0.13)	0.10 (0.19)	0.27** (0.10)
Source Z-Score	X	X	X	X	X
Clusters	60	60	59	60	60
Num.Obs.	17390	18555	9491	3780	12671
R2	0.017	0.022	0.027	0.033	0.012
R2 Adj.	0.017	0.021	0.026	0.031	0.012

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

All models use a source-level Z-Score normalization with the optimal bandwidth. Each column restricts to the indicated topic category. † Standard errors computed via wild cluster restricted (WCR) bootstrap (Cameron, Gelbach & Miller 2008) with Webb six-point weights (B=99999, null imposed), clustered by source domain. ‡ Cluster-robust SEs from rdrobust.

significant for economic coverage (0.29, $p < 0.05$) and other coverage (0.27, $p < 0.01$), and positive but imprecise for political (0.155), military (0.012), and cultural (0.10) topics. That the divergence is largest in economic coverage is suggestive, though we do not theorize topic-level variation: states in our sample maintained substantial trade relationships with both Russia and the West, and economic framing may be the domain where strategic management of coverage is most consequential. The within-topic results also indicate that the H2 finding is not an artifact of compositional change—the divergence between state-owned and independent media is present within topic categories, not merely across them. Leave-one-out results and additional fixed-effects specifications are reported in Tables A3–A6.

H3: Russia-Aligned State Media Responds More Negatively than Neutral State Media

The central test of our argument compares state-owned media across security relationships. A domestic audience management account predicts that Russia-aligned states suppress negative coverage of their security patron, producing a smaller decline in sentiment than neutral states. Our signaling account predicts the opposite: because formal

channels of dissent are foreclosed by the alliance constraint, Russia-aligned states use state media to signal distance from the patron to external audiences, producing a larger decline than neutral states whose formal channels remain open.

Table 4 presents the results. Models 1 and 2 report separate RDD estimates for Russia-aligned and neutral state media, restricted to non-military articles. Russia-aligned state media exhibits a significant decline of 0.220 within-source standard deviations ($p < 0.05$). Neutral state media, by contrast, shows no significant change (0.055, $p > 0.1$). The pattern is striking: it is the constrained states—those with formal security agreements with Russia, the same states that refused to condemn the invasion through diplomatic channels—whose state media shifted most negatively.

Models 3 through 5 estimate the RDID interaction directly. The Russia-aligned $\times$ treatment coefficient is negative across all specifications, indicating that Russia-aligned state media responded more negatively than neutral state media. Model 3, estimated on the full sample at the optimal bandwidth, yields an interaction of -0.19 that is not statistically significant. However, once we omit military articles (Model 4), the estimated coefficient increases in magnitude to -0.33 and becomes statistically significant ($p < 0.05$). Model 5 applies coarsened exact matching on publication date and topic to the non-military sample, producing an estimate of -0.38 ($p < 0.05$). The matched specification is our preferred estimate for this comparison: with only 13 source-level clusters, ensuring compositional equivalence across groups is particularly important, and matching on observables addresses the concern that the two groups differ in the timing or subject matter of their coverage. Inference for all RDID models uses the wild cluster restricted bootstrap with Webb six-point weights to account for the small number of clusters. A fully disaggregated analysis estimating separate RDDs within each source-topic cell, aggregated via both a pairs bootstrap and a Bayesian bootstrap, produces consistent results (Table A11).

Figure 7 presents the discontinuity by alignment group. Russia-aligned state media (blue pre-invasion, red post-invasion) exhibits a visible downward shift at the cutoff, while neutral state media (green pre-invasion, orange post-invasion) remains essentially

flat. The divergence at the cutoff is the RDID estimate. The near-zero (neutral) pre-invasion level for Russia-aligned state media may reflect the tight control exercised over these outlets—Azerbaijan’s and Kazakhstan’s daily means cluster at exactly zero before the invasion (Figure 2)—which makes the post-invasion shift into negative territory even more notable. The null result for neutral state media is not a floor effect: these outlets begin well above the scale floor yet barely move, unlike independent outlets in the same countries (Appendix Figure A8).

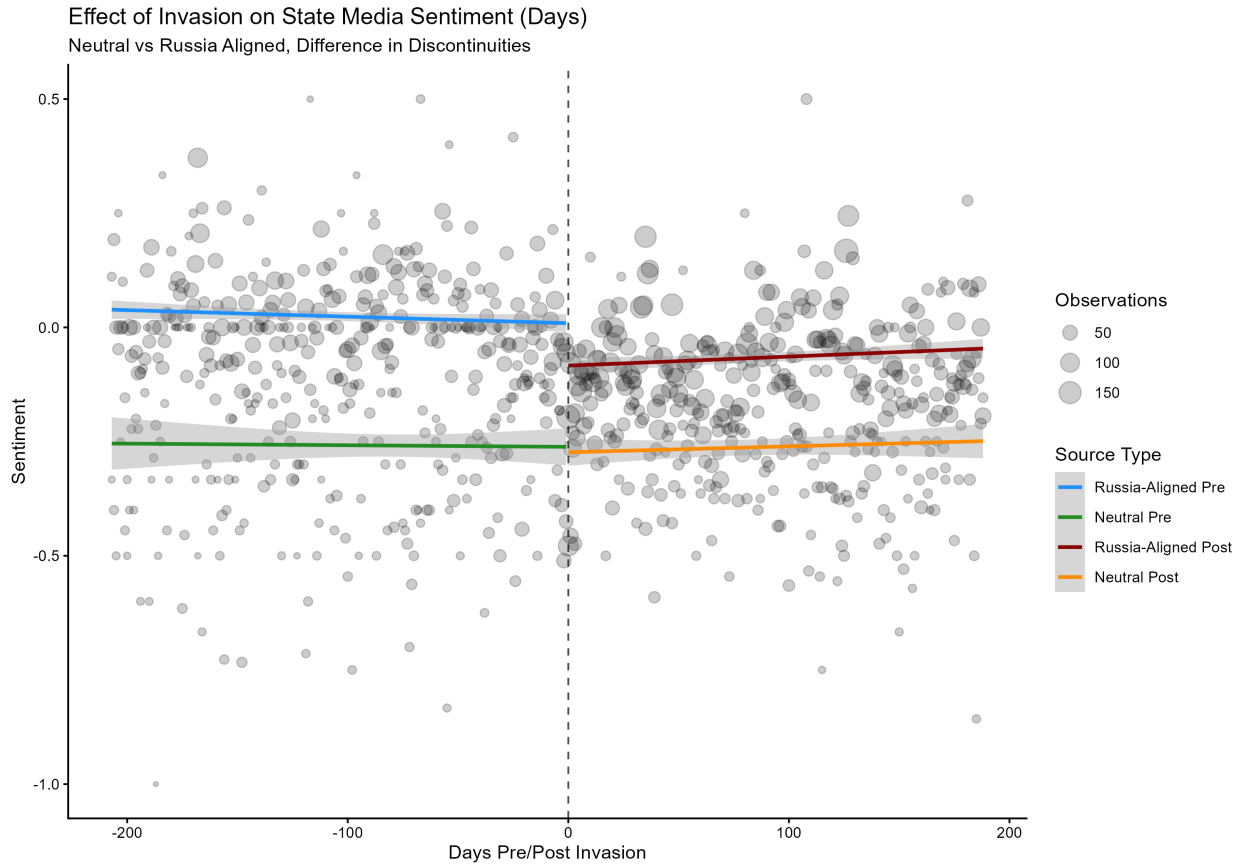


Figure 7: RDD estimates of the effect of the invasion on sentiment toward Russia in state-owned media, by security alignment. Sample restricted to state-owned media in non-NATO countries. Each point represents one day, sized by the number of articles published. The dashed line marks February 24th, 2022.

The within-outlet shift in Armenia’s state broadcaster illustrates the mechanism and shows it operating through selection rather than editorial voice. In January, the English-language service of Public Radio of Armenia carried “Opening of regional communications to give Armenia rail access to Russia, Europe, Iran, China,” a report on

Table 4: H3a Alignment Interaction — Z-Score (Probabilistic Time)

	(1) CSTO‡	(2) Neutral‡	(3)†	(4)†	(5)†
Invasion	-0.220*	0.055	-0.068	0.098	0.098
	(0.099)	(0.101)	(0.080)	(0.070)	(0.069)
CSTO x Treatment			-0.19	-0.33*	-0.38*
			(0.11)	(0.11)	(0.11)
Source Z-Score	X	X	X	X	X
Clusters	9	4	13	13	13
Num.Obs.	17733	4802	7164	6094	5721
R2			0.025	0.015	0.014
R2 Adj.			0.024	0.013	0.013

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

All models use a source-level Z-Score normalization. Sample restricted to state-owned media in non-NATO countries. Models 1-2 show separate RDD estimates for CSTO-aligned and neutral countries, filtered to omit military topics. Models 3-5 estimate the CSTO x treatment interaction. Model 3 uses the optimal bandwidth. Model 4 omits military topics. Model 5 uses exact matching on topic and date of publication for the non-military data. † Standard errors computed via wild cluster restricted (WCR) bootstrap (Cameron, Gelbach & Miller 2008) with Webb six-point weights (B=99999, null imposed), clustered by source domain. ‡ Cluster-robust SEs from rdrobust.

the prime minister’s infrastructure plans in which Russia figured as one economic partner among several—coverage the classifier coded positive toward Russia.¹² By March, the same outlet was running “Russia walks out of Council of Europe,” foregrounding Russia’s rupture with the European institutions of which Armenia is itself a member—coverage coded negative. Neither article is editorial commentary in the outlet’s own voice; the March piece is largely wire-style reporting of Russia’s withdrawal and its combative justification. What changed is the orientation of the Russia coverage the outlet selected and foregrounded, from economic partnership to institutional rupture—and because it is a state outlet, that change in what it carries is readable as a deliberate choice rather than editorial drift. The shift occurred while Armenia withheld formal condemnation, abstaining on the UN General Assembly resolution against the invasion: the media-versus-diplomacy gap our account predicts for constrained states, captured in a single outlet.

¹²Translations are machine-generated and reproduce the title and opening supplied to the classifier; the source is en.armradio.am.

The direction of this result provides leverage for distinguishing between the competing theoretical accounts. A domestic audience management account and our signaling account generate opposite predictions for hypothesis 3, and the data are consistent with the signaling account. Russia-aligned states did not suppress negative coverage to protect the alliance relationship; they permitted—or directed—a sharper negative shift in their state media than neutral states whose formal channels were unconstrained. As discussed in the Interpreting the RDID section, we interpret this differential as conditional on group membership rather than as a causal effect of Russia-alignment per se. Two features of the analysis—a sensitivity analysis bounding the role of unobserved factors and an alternative group partition—bear on whether the pattern reflects the mechanism our theory describes rather than an omitted confounder; we present both in the Robustness section below.

Robustness

We subject our results to a series of robustness checks organized around the principal threats to identification. We begin with a placebo test of the continuity assumption. We then present our sensitivity analysis, which we generalize as the matched sensitivity procedure for difference-in-discontinuities, a method for bounding unobserved confounding in cross-group discontinuity comparisons that lack the structure required for causal identification. We then examine the alternative group partition discussed in the Case section. We close by assessing our use of topic fixed effects and summarizing the remaining checks, whose full results are reported in the appendix.

Placebo Test

The most plausible threat to continuity in our setting is differential anticipation—sentiment may have begun shifting before the invasion as its likelihood became clear, and countries aligned with Russia may have had inside knowledge. A placebo cutoff at December 7th, 2021—the date the United States publicly announced that Russia intended to invade Ukraine—targets this threat at the point where it should be most

visible. If media sentiment were already shifting discontinuously in the weeks before the invasion, we would expect to detect a significant break at this date. We find no significant discontinuity in the full sample (Appendix Table A12) or in the state media of Russia-aligned countries specifically—the group for which differential anticipation would matter (Appendix Table A13). The only individually significant break in that group, Belarus’s, runs in the positive direction, as discussed in the Research Design section. These results are consistent with the interpretation that the invasion itself, rather than its anticipation, produced the sharp shift in coverage.

Sensitivity to Unobserved Confounding

To our knowledge, cross-group RDID comparisons like ours are not routinely subjected to formal sensitivity analysis; we propose the matched sensitivity procedure for difference-in-discontinuities—coarsened exact matching to address observable compositional differences across groups, paired with the Cinelli and Hazlett (2020) framework to bound the role of unobservables—as a standard robustness test for the setting. Establishing such tests for associational comparisons is itself a contribution: as Lieberman (2016) argues, the descriptive and correlational stages of inquiry merit the same methodological care as causal ones, and a differential shown robust to this scrutiny is a firmer foundation for the causal work that might follow.

The central threat to our RDID estimates is that an unobserved factor correlated with Russia-alignment produces a differential jump in sentiment at the cutoff through a channel other than the one our theory describes. We address this by combining coarsened exact matching with the sensitivity framework of Cinelli and Hazlett (2020).

We first construct a matched sample using coarsened exact matching (CEM; Iacus et al., 2012) on publication date and topic, pairing Russia-aligned and neutral articles that were published on the same day and assigned to the same topic category. Matching on these observable characteristics ensures that the RDID comparison is not driven by differences in the timing or subject matter of coverage across groups. We then estimate the RDID on the matched sample using the same specification as our

main analysis (Equation 3), with the z-scored outcome, topic fixed effects, and the fully interacted running variable, treatment, and group indicator. The Cinelli and Hazlett (2020) framework, developed for OLS coefficients, applies directly to the RDID interaction coefficient ($T_p \times G_p$) in this specification.

We apply `sensemkr` to the matched-sample regression, designating the RDID interaction as the coefficient of interest, and use this confounding framework as a disciplined stress-test of the association rather than as a claim that the differential is a causal effect of group membership. The robustness value is 1.9%: an unobserved factor would need to explain at least 1.9% of the residual variance of both group assignment and sentiment to drive the estimated differential to zero.¹³ To calibrate this, we benchmark against the invasion indicator itself—the strongest substantive predictor in the design. Even granted a near-mechanical 56% association with the Russia-aligned $\times$ treatment interaction, an unobserved factor would still need to be 1.5 times as strongly associated with sentiment as the invasion itself to drive the differential to zero.¹⁴ Figure 8 presents the contour plot, with the dashed line marking the invasion indicator’s own partial R^2 with sentiment to calibrate the vertical scale. Because our source fixed effects absorb all time-invariant outlet and country characteristics, the confounders that could satisfy this joint condition form a narrow class: they would have to vary differentially at the cutoff by group.

¹³We compute the robustness value using bootstrapped standard errors from the WCR bootstrap rather than the OLS standard errors from `lm()`, which do not account for clustering. The OLS-based robustness value would overstate the degree of robustness.

¹⁴The main treatment indicator is a component of the interaction $T_p \times G_p$, so at full strength the benchmark is degenerate: $k_d = 1$ implies a confounder partial R^2 with the interaction of at least one, which the procedure cannot evaluate. We therefore set $k_d = 0.25$; even at a quarter of the benchmark’s strength the implied confounder must explain 56% of the interaction’s residual variance. The partial R^2 with sentiment required to drive the differential to zero at that point is 0.0015, against the invasion indicator’s own 0.0010.

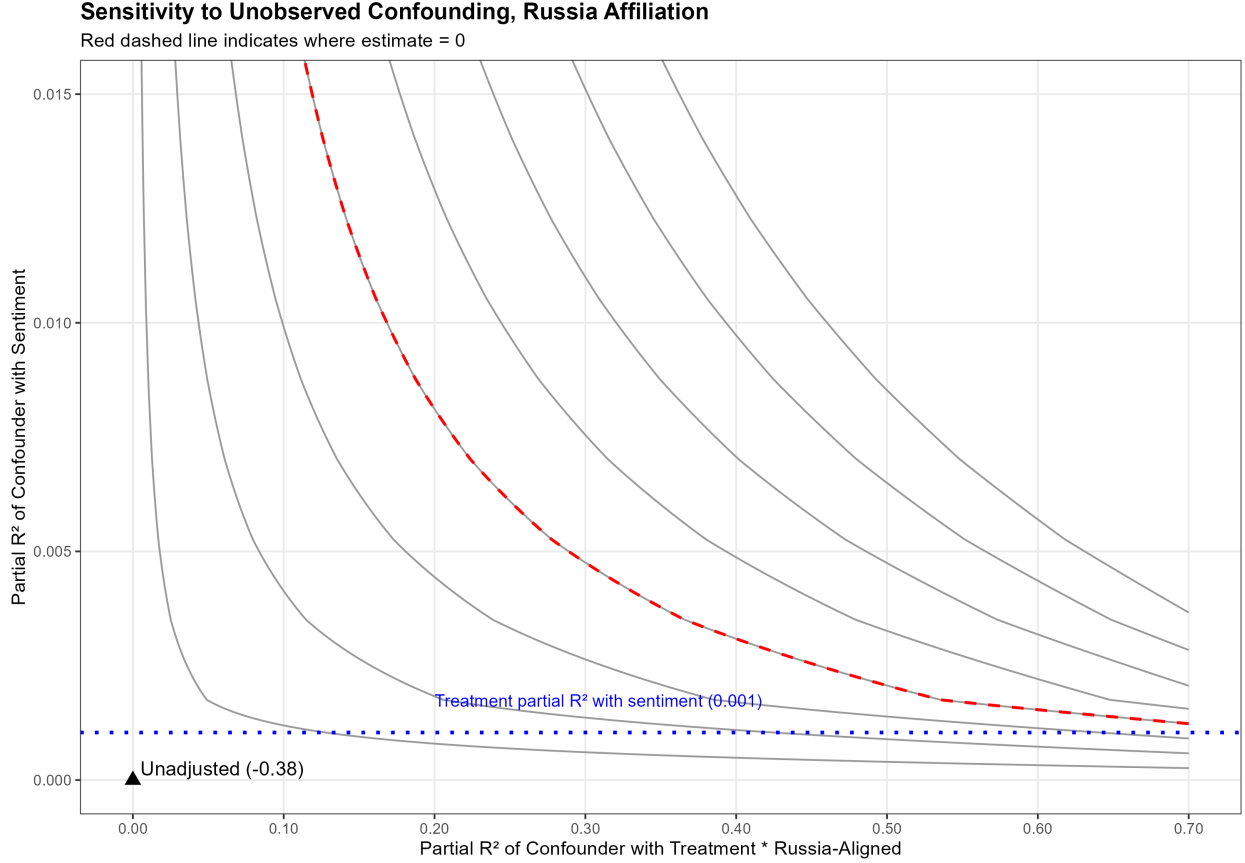


Figure 8: Sensitivity of the Russia-aligned $\times$ treatment interaction to unobserved confounding, following Cinelli and Hazlett (2020). The red dashed line indicates the combinations of confounder strength that would drive the estimated differential to zero. The triangle marks the unadjusted estimate; the dashed horizontal line marks the invasion indicator’s own partial R^2 with sentiment (≈ 0.001), calibrating the vertical scale.

An Alternative Group Partition

We test an alternative group partition that provides a direct check on a prominent competing explanation. If the differential response were driven by exposure to the invasion’s stated pretext or by cultural proximity to Russia rather than by the alliance constraint, we would expect a similar pattern when states are partitioned by the share of ethnic Russian population—the dimension Russia itself invoked to justify the invasion. Table 5 presents the cross-classification of countries by Russia-alignment and ethnic Russian population share. The two dimensions cross-cut cleanly: Azerbaijan and Armenia are Russia-aligned but have low ethnic Russian populations, while Moldova and Georgia are neutral but have high ethnic Russian populations. This cross-cutting structure gives

Table 5: Ethnic Russian Population x Russia Security Alignment. Country Classification

	Low Alignment (Neutral)	High Alignment (CSTO)
Low Ethnic	Kosovo, Serbia	Azerbaijan, Armenia
High Ethnic	Moldova, Georgia	Belarus, Kazakhstan

the test discriminating power: the two explanations make divergent predictions, and a result driven by pretext exposure or cultural proximity would appear in the ethnic Russian partition regardless of Russia-alignment. Because the partition is orthogonal to alignment, a result that loads on alignment but not on ethnic-Russian share isolates our proposed mechanism rather than a correlate of it.

Table 6 presents the results. Models 1 and 2 report separate RDD estimates for high and low ethnic Russian population countries; neither shows a significant discontinuity. Models 3 through 5 estimate the interaction directly. The high Russian population $\times$ treatment coefficient is substantively small and statistically insignificant across all specifications, ranging from -0.055 to 0.015 . The contrast with the Russia-aligned partition is stark: the same sample, the same methodology, and the same estimation procedure produce a significant differential when countries are split by security alignment and no differential when split by ethnic composition. This pattern provides no support for the pretext-exposure or cultural proximity explanations and is consistent with a mechanism that operates through the alliance constraint rather than through exposure or ethnic ties to Russia. Leave-one-out results and additional specifications are reported in Tables A7–A10.

Topic Adjustment and the Total Effect

As discussed in the Identification section, our topic-adjusted estimates identify a controlled direct effect rather than the total effect of the invasion. For transparency, we report total-effect estimates without topic adjustment in Appendix Tables A14–A16. For all three hypotheses, the total effect is substantively similar to its topic-adjusted

Table 6: H3b Ethnic Russian Population — Null Result, Z-Score (Probabilistic Time)

	(1) High‡	(2) Low‡	(3)†	(4)†	(5)†
Invasion	-0.079 (0.066)	-0.108 (0.102)	-0.17 (0.10)	-0.13 (0.10)	-0.143 (0.099)
High Russian Pop x Treatment			-0.055 (0.133)	-0.0024 (0.1479)	0.015 (0.145)
Source Z-Score	X	X	X	X	X
Clusters	9	4	13	13	13
Num.Obs.	13942	8593	7202	6092	5992
R2			0.022	0.012	0.011
R2 Adj.			0.021	0.010	0.010

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

All models use a source-level Z-Score normalization. Models 1-2 show separate RDD estimates for high and low ethnic Russian population countries. Models 3-5 estimate the high Russian population x treatment interaction. Model 3 uses the optimal bandwidth. Model 4 omits military topics. Model 5 uses matched data. † Standard errors computed via wild cluster restricted (WCR) bootstrap (Cameron, Gelbach & Miller 2008) with Webb six-point weights (B=99999, null imposed), clustered by source domain. ‡ Cluster-robust SEs from rdrobust.

counterpart.¹⁵ Estimated on the main samples, the versions without topic fixed effects remain statistically significant and close to the adjusted estimates (0.208 versus 0.222 for hypothesis 2; -0.37 versus -0.33 for hypothesis 3). To address concerns about post-treatment confounding directly, we also estimate variants of our main results that use only pre-invasion topic and source-level adjustments. These variants approximate the sequential g-estimation approach proposed in Acharya et al. (2016), foreclosing post-treatment conditioning by construction, and the estimated coefficients under this construction are similar in magnitude and maintain statistical significance (Appendix Tables A17 and A18).

Additional Robustness

We assess whether our results are driven by any single country or outlet through leave-one-out analysis, sequentially dropping each country and each source; results are stable

¹⁵Direct comparisons to our main models for hypotheses 2 and 3 are complicated. Both main specifications omit military-related articles, so a total effect estimated on all articles uses meaningfully different data. Tables A15 and A16 therefore report the like-for-like comparisons for each sample (both with and without military-related articles), including the main results with topic fixed effects removed.

across all exclusions and are reported in the appendix. The matched estimates presented in the main text (Tables 2 and 4) are consistent with the unmatched specifications, indicating that observable differences in article timing and subject matter across groups do not account for the differential response. For hypothesis 2, restricting the sample to non-NATO countries does not attenuate the result (Table 2). Omitting military articles and re-estimating all specifications leaves the results intact across all hypotheses (Tables 2 and 4), and estimating the RDID within each topic category separately (Table 3) indicates that the differential response is present within non-military topics. Clustering at the country level instead of the source level leaves the preferred specifications for hypotheses 2 and 3 statistically significant ($p = 0.021$ and $p = 0.032$).¹⁶ Alternative forcing variable constructions (publication date rather than date-time; dropping articles with imputed timestamps) are reported in the Alternative Forcing Variable Constructions appendix, and bandwidth variation (results stable from 0.5 to 1.5 times the optimal bandwidth) is reported alongside each hypothesis’s additional specifications in the appendix.

Conclusion

The Russian invasion of Ukraine confronted Russia’s security partners with a dilemma. These states had genuine reasons to oppose the invasion, but the directly attributable channels through which the discipline has primarily studied dissent—UN votes, official statements, public breaks—risked provoking the very patron that guarantees their security. Our findings show that these states turned to state-owned media. The invasion produced a sharp negative shift in sentiment toward Russia across the region (H1), and state-owned media managed this shift more carefully than independent media, consistent with strategic control of coverage (H2). That independent media responded more strongly than state media is itself evidence that state outlets operate under political

¹⁶We cluster at the source level in the main analysis because it matches the level at which we standardize the outcome and preserves more clusters for the bootstrap (see the Estimation and Inference section). Country-level clustering leaves 11 clusters for hypothesis 2 and 8 for hypothesis 3. We use the same wild cluster restricted bootstrap with Webb weights as in the main analysis (MacKinnon & Webb, 2017), which is designed for cluster counts this small.

constraints that shape their coverage—if both responded identically, there would be no strategic management to detect. The central finding, however, is that state-owned media in Russia-aligned states shifted more negatively than state-owned media in neutral states (H3)—the opposite of what a domestic audience management account predicts, and precisely the pattern our signaling argument describes. States whose formal channels were most constrained produced the sharpest media signal, despite these same states refusing to condemn the invasion through formal diplomatic channels. An alternative partition by ethnic Russian population share produced no differential, ruling out cultural proximity as a driver and providing further evidence that the mechanism operates through the alliance constraint.

These findings speak to a long-standing disconnect between international relations scholarship and intelligence practice. Western intelligence services have monitored state media as a window into regime preferences since the Cold War, treating shifts in tone and emphasis as indicators of factional dynamics and policy direction within regimes whose formal diplomatic output is tightly controlled. The discipline, by contrast, has studied state media almost exclusively as a tool of domestic audience management. Our analysis demonstrates that the signals practitioners monitor are not idiosyncratic to particular crises or analysts—they are theoretically structured and empirically identifiable. The patterns we document follow directly from the logic of alliance constraints, and they discriminate between competing theoretical accounts: the domestic management and signaling accounts generate opposite predictions for how Russia-aligned state media should respond to the invasion, and the data are consistent with the signaling account alone. State media sentiment, in short, is not noise. It is a legible instrument of international communication, one that the discipline can study with the same rigor it applies to formal diplomatic behavior.

Our second contribution identifies the channel through which this communication operates. The signaling literature has focused on how states communicate resolve or commitment through directly attributable instruments—UN votes, official statements, military deployments—precisely because their visibility generates the costs that make

them credible. We show that when these channels are foreclosed by alliance constraints, states turn to institutionally mediated ones. State media has just this character: tied closely enough to the government that a shift in its coverage reads as intentional, but not so formal that the shift becomes an act of state. The signal is costly enough to shift the receiver’s beliefs about the sender’s alignment, but institutionally buffered—through the structural separation between government and editorial function—to avoid crossing the threshold that has historically triggered patron retaliation. This logic is not unique to the present case. Any setting in which a subordinate actor depends on a patron but must manage relationships with other audiences creates pressure for signaling through channels that are observable to the intended audience but do not commit the sender to an official position. It should extend beyond international relations to any patron-client hierarchy—subnational governments constrained by central authorities, or corporate subsidiaries navigating the preferences of a parent firm—though we leave these extensions to future work. The same pressure is strongest where patron and client are mutually dependent—whether through security commitments, economic ties, or both—such that the patron’s cost of punishing media-level dissent exceeds the benefit of enforcing message discipline.

Within international relations, natural extensions include states in China’s security orbit today, US-backed regimes in Central America during the 1980s and 1990s, where similar dynamics of patron dependence and constrained dissent shaped the relationship between state media and official policy, and contemporary cases in which US allies navigate the tension between security dependence and domestic or international pressure regarding conflicts such as the war in Gaza. The 2022 invasion provides unusual analytical clarity—a discontinuity, a well-defined alliance structure, and competing theoretical predictions—but the underlying mechanism should operate wherever directly attributable channels of dissent are foreclosed by dependence on a powerful patron.

Methodologically, our approach offers two contributions. First, the pairing of prompted large language model classification with a regression discontinuity design provides a template for large-scale analysis of media signals in contexts where hand-coding is in-

feasible. The prompted LLM approach is not incidental to the measurement strategy: by specifying sentiment toward Russia as the classification target, we measure the article’s orientation toward a specific actor rather than the overall tone of the coverage, a distinction that off-the-shelf sentiment classifiers cannot make. Second, our matched sensitivity procedure for difference-in-discontinuities—coarsened exact matching combined with the sensitivity framework of Cinelli and Hazlett (2020)—offers a transparent, reusable way to assess how robust a cross-group comparison is to omitted factors when group membership is observational rather than assigned, a common setting in comparative research that is rarely subjected to formal sensitivity analysis. Disciplining associational comparisons in this way is, following Lieberman (2016), valuable in its own right: it strengthens the descriptive and correlational foundations on which later causal work depends.

Three limitations merit discussion. First, our findings are drawn from a single crisis. The invasion of Ukraine provides unusual analytical clarity, but the generalizability of the results depends on whether the same logic operates in other settings where directly attributable channels are constrained by patron dependence. Testing this across other alliance structures is a priority for future work. Second, we document sender behavior but not receiver response. The institutional infrastructure for monitoring state media is well-established—Western intelligence services have done so systematically for decades—and states in our sample operate in an environment where they can reasonably expect their coverage to be observed. But we do not trace the signal from sender to receiver, and we cannot confirm that the shifts we identify were detected, interpreted, or acted upon by the intended audience. Third, as we discuss in the theory section, our data cannot distinguish whether the state media shifts reflect top-down strategic direction by regime leadership or the expression of elite factional pressure through outlets that different factions influence. Both mechanisms produce the same observable pattern, and the divergence itself—that the regime is not unified in its public posture toward its security patron—is analytically meaningful regardless of which internal process produces it.

The sender-receiver gap points toward productive avenues for future research. Two approaches could help close the loop. First, examining whether Western sanctions targeted Russia-aligned states differentially based on their media posture would provide evidence on whether the signals we identify correlate with policy responses. Second, analyzing whether publicly traded firms in Russia-aligned countries with deep Western ties experienced financial penalties correlated with state media sentiment shifts would test whether markets—another audience that monitors public information—responded to the same signals. Neither measure perfectly captures signal receipt, but both would move the analysis from documenting the sender’s behavior to tracing its consequences. More broadly, the theoretical framework developed here provides a basis for interpreting what intelligence practitioners have long observed. Western governments have monitored state media as a source of regime preferences for decades; this paper offers a theoretical account of why the practice works. The signals are not ad hoc—they follow from the structure of alliance constraints, which means they can be interpreted more systematically and, as we have demonstrated, quantified at scale. As the international system moves toward a more multipolar order, patron-client alliances will continue to press the communication bind upon smaller states. Understanding how they communicate under these conditions will be a fundamental component of studying international relations in the decades ahead.

References

- Abadie, A., Athey, S., Imbens, G. W., & Wooldridge, J. M. (2022). When Should You Adjust Standard Errors for Clustering? *The Quarterly Journal of Economics*, 138(1), 1–35. <https://doi.org/10.1093/qje/qjac038>
- Acharya, A., Blackwell, M., & Sen, M. (2016). Explaining Causal Findings Without Bias: Detecting and Assessing Direct Effects. *American Political Science Review*, 110(3), 512–529. <https://doi.org/10.1017/S0003055416000216>
- Allison, R. (2008). Russia resurgent? Moscow’s campaign to ‘coerce Georgia to peace’. *International Affairs*, 84(6), 1145–1171. <https://doi.org/10.1111/j.1468-2346.2008.00762.x>
- Aronow, P. M., & Samii, C. (2016). Does Regression Produce Representative Estimates of Causal Effects? *American Journal of Political Science*, 60(1), 250–267. <https://doi.org/10.1111/ajps.12185>
- Banco, E., Graff, G., Seligman, L., Toosi, N., & Ward, A. (2023). ‘Something Was Badly Wrong’: When Washington Realized Russia Was Actually Invading Ukraine. *Politico*. Retrieved June 1, 2026, from <https://www.politico.com/news/magazine/2023/02/24/russia-ukraine-war-oral-history-00083757?nlid=630318>
- Bank, W. (2021). World Integrated Trade Solution (WITS). <https://wits.worldbank.org/#>
- Bansak, K., & Nowacki, T. (2022, June). Effect Heterogeneity and the Challenges of Causal Attribution in Regression Discontinuity Designs. <https://doi.org/10.31235/osf.io/vj34m>
- BBC Monitoring. (2016, September). *Written Evidence to the House of Commons Defence Committee (BBC0002)* (Government Report). House of Commons. Retrieved March 18, 2026, from <https://committees.parliament.uk/writtenevidence/72975/html/>

- Blankenship, B., & Lin-Greenberg, E. (2022). Trivial Tripwires?: Military Capabilities and Alliance Reassurance. *Security Studies*, 31(1), 92–117. <https://doi.org/10.1080/09636412.2022.2038662>
- Bueno de Mesquita, B. (Ed.). (2003). *The logic of political survival*. MIT Press.
- Calonico, S., Cattaneo, M. D., & Farrell, M. H. (2020). Optimal bandwidth choice for robust bias-corrected inference in regression discontinuity designs. *The Econometrics Journal*, 23(2), 192–210. <https://doi.org/10.1093/ectj/utz022>
- Cameron, A. C., Gelbach, J. B., & Miller, D. L. (2008). Bootstrap-Based Improvements for Inference with Clustered Errors. *The Review of Economics and Statistics*, 90(3), 414–427. Retrieved March 20, 2024, from <https://ideas.repec.org/a/tpr/restat/v90y2008i3p414-427.html>
- Cameron, A. C., & Miller, D. L. (2015). A Practitioner’s Guide to Cluster-Robust Inference. *Journal of Human Resources*, 50(2), 317–372. <https://doi.org/10.3368/jhr.50.2.317>
- Cattaneo, M. D., Idrobo, N., & Titiunik, R. (2024, April). *A Practical Introduction to Regression Discontinuity Designs: Extensions* (1st ed.). Cambridge University Press. <https://doi.org/10.1017/9781009441896>
- Cattaneo, M. D., Idrobo, N., & Titiunik, R. (2019, November). *A Practical Introduction to Regression Discontinuity Designs: Foundations* (1st ed.). Cambridge University Press. <https://doi.org/10.1017/9781108684606>
- Cattaneo, M. D., Keele, L., & Titiunik, R. (2023, March). Covariate Adjustment in Regression Discontinuity Designs. In *Handbook of Matching and Weighting Adjustments for Causal Inference* (1st ed., pp. 153–168). Chapman; Hall/CRC. <https://doi.org/10.1201/9781003102670-8>

- Cattaneo, M. D., & Titiunik, R. (2022). Regression Discontinuity Designs. *Annual Review of Economics*, 14(1), 821–851. <https://doi.org/10.1146/annurev-economics-051520-021409>
- Cheterian, V. (2009). The August 2008 war in Georgia: From ethnic conflict to border wars. *Central Asian Survey*, 28(2), 155–170. <https://doi.org/10.1080/02634930903056768>
- Chong, D., & Druckman, J. N. (2007). Framing Public Opinion in Competitive Democracies. *American Political Science Review*, 101(4), 637–655. <https://doi.org/10.1017/S0003055407070554>
- Cinelli, C., & Hazlett, C. (2020). Making Sense of Sensitivity: Extending Omitted Variable Bias. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 82(1), 39–67. <https://doi.org/10.1111/rssb.12348>
- Cohen, R. (1987). *Theatre of power: The art of diplomatic signalling*. Longman.
- Dylan, H., & Maguire, T. J. (2022). Secret Intelligence and Public Diplomacy in the Ukraine War. *Survival*, 64(4), 33–74. <https://doi.org/10.1080/00396338.2022.2103257>
- Egorov, G., Guriev, S., & Sonin, K. (2009). Why Resource-poor Dictators Allow Freer Media: A Theory and Evidence from Panel Data. *American Political Science Review*, 103(4), 645–668. <https://doi.org/10.1017/S0003055409990219>
- Enikolopov, R., Petrova, M., & Zhuravskaya, E. (2011). Media and Political Persuasion: Evidence from Russia. *American Economic Review*, 101(7), 3253–3285. <https://doi.org/10.1257/aer.101.7.3253>
- Entman, R. M. (2004). *Projections of power: Framing news, public opinion, and U.S. foreign policy* (Nachdr.). University of Chicago Press.
- Favere, A. (2024). Debating Open Source: A Practitioner’s Perspective. *Studies in Intelligence*, 68(3). Retrieved March 18, 2026, from <https://www.cia.>

gov / resources / csi / studies - in - intelligence / studies - in - intelligence - 68 - no - 3 - extracts - september - 2024 / debating - open - source - a - practitioners - perspective /

- Fearon, J. D. (1997). Signaling Foreign Policy Interests: Tying Hands versus Sinking Costs. *Journal of Conflict Resolution*, 41(1), 68–90. <https://doi.org/10.1177/0022002797041001004>
- Flamer, N. (2023). ‘The enemy teaches us how to operate’: Palestinian Hamas use of open source intelligence (OSINT) in its intelligence warfare against Israel (1987-2012). *Intelligence and National Security*, 38(7), 1171–1188. <https://doi.org/10.1080/02684527.2023.2212556>
- Fu, J., & Slough, T. (2026). Heterogeneous Treatment Effects and Causal Mechanisms. *American Political Science Review*, 1–18. <https://doi.org/10.1017/S0003055426101580>
- Fuhrmann, M., & Sechser, T. S. (2014). Signaling Alliance Commitments: Hand-Tying and Sunk Costs in Extended Nuclear Deterrence. *American Journal of Political Science*, 58(4), 919–935. <https://doi.org/10.1111/ajps.12082>
- Gandhi, J. (2008). *Political institutions under dictatorship*. Cambridge University Press.
- Gehlbach, S., & Sonin, K. (2014). Government control of the media. *Journal of Public Economics*, 118, 163–171. <https://doi.org/10.1016/j.jpubeco.2014.06.004>
- Gershkoff, A., & Kushner, S. (2005). Shaping Public Opinion: The 9/11-Iraq Connection in the Bush Administration’s Rhetoric. *Perspectives on Politics*, 3(3), 525–537. <https://doi.org/10.1017/S1537592705050334>
- Gong, L., Hopkins, D. J., & Wolken, S. (2026). The Limits of De-Politicizing—and Also of Annotation: A Case Study in Russian Media Outlets’ Social Media Posts, 2016–2024. *Proceedings of the International AAAI Conference on*

- Web and Social Media*, 20(1), 889–909. <https://doi.org/10.1609/icwsm.v20i1.42672>
- Götz, E., & Staun, J. (2022). Why Russia attacked Ukraine: Strategic culture and radicalized narratives. *Contemporary Security Policy*, 43(3), 482–497. <https://doi.org/10.1080/13523260.2022.2082633>
- Grembi, V., Nannicini, T., & Troiano, U. (2016). Do Fiscal Rules Matter? *American Economic Journal: Applied Economics*, 8(3), 1–30. <https://doi.org/10.1257/app.20150076>
- Guriev, S. M., & Treisman, D. (2022). *Spin dictators: The changing face of tyranny in the 21st century*. Princeton university press.
- Hausman, C., & Rapson, D. S. (2018). Regression Discontinuity in Time: Considerations for Empirical Applications. *Annual Review of Resource Economics*, 10(1), 533–552. <https://doi.org/10.1146/annurev-resource-121517-033306>
- Hirschman, A. O. (1970). *Exit, voice and loyalty: Responses to decline in firms, organisations and States*. Harvard university press.
- Huseynov, V. (2022). Azerbaijan and Russia Sign Declaration on Allied Cooperation. *Foreign Policy*, 19(25). <https://jamestown.org/azerbaijan-and-russia-sign-declaration-on-allied-cooperation/>
- Iacus, S. M., King, G., & Porro, G. (2012). Causal Inference without Balance Checking: Coarsened Exact Matching. *Political Analysis*, 20(1), 1–24. <https://doi.org/10.1093/pan/mpr013>
- Iyengar, S., & Kinder, D. R. (2010). *News that matters: Television and American opinion* (Updated ed). University of Chicago Press.
- Jazeera, A. (2022, January). US and NATO deliver responses to Russian demands over Ukraine. Retrieved April 28, 2024, from <https://www.aljazeera.com/>

[news/2022/1/26/us-nato-deliver-responses-to-russian-demands-over-ukraine](#)

- Kassing, J. W. (1997). Articulating, antagonizing, and displacing: A model of employee dissent. *Communication Studies*, 48(4), 311–332. <https://doi.org/10.1080/10510979709368510>
- Lieberman, E. S. (2016). Can the Biomedical Research Cycle be a Model for Political Science? *Perspectives on Politics*, 14(4), 1054–1066. <https://doi.org/10.1017/S153759271600298X>
- Lovell, M. C. (1963). Seasonal Adjustment of Economic Time Series and Multiple Regression Analysis. *Journal of the American Statistical Association*, 58(304), 993–1010. <https://doi.org/10.1080/01621459.1963.10480682>
- MacKinnon, J. G., & Webb, M. D. (2017). Wild Bootstrap Inference for Wildly Different Cluster Sizes. *Journal of Applied Econometrics*, 32(2), 233–254. <https://doi.org/10.1002/jae.2508>
- Malesky, E., & Schuler, P. (2010). Nodding or Needling: Analyzing Delegate Responsiveness in an Authoritarian Parliament. *American Political Science Review*, 104(3), 482–502. <https://doi.org/10.1017/S0003055410000250>
- Mankoff, J. (2022). Turkey’s Balancing Act on Ukraine Is Becoming More Precarious. *Foreign Policy*. <https://foreignpolicy.com/2022/03/10/turkey-ukraine-russia-war-nato-erdogan/>
- Manor, I. (2019). *The Digitalization of Public Diplomacy*. Palgrave Macmillan US.
- McCombs, M. (2002). The agenda-setting role of the mass media in the shaping of public opinion. <http://sticerd.lse.ac.uk/dps/extra/McCombs>
- Montgomery, J. M., Nyhan, B., & Torres, M. (2018). How Conditioning on Post-treatment Variables Can Ruin Your Experiment and What to Do about

- It. *American Journal of Political Science*, 62(3), 760–775. <https://doi.org/10.1111/ajps.12357>
- Moratz, D., Springman, J., Wibbels, E., Adiguzel, F. S. S., Villamizar Chaparro, S. M., Lin, Z.-R., Romero, D., Soltani, M., Su, H., & Swami, J. (2025, August). Tracking Civic Space in Developing Countries with a High-Quality Corpus of Domestic Media and Transformer Models. https://doi.org/10.31235/osf.io/zp3sr_v2
- Morrow, J. D. (1991). Alliances and Asymmetry: An Alternative to the Capability Aggregation Model of Alliances. *American Journal of Political Science*, 35(4), 904. <https://doi.org/10.2307/2111499>
- Nations, U. (1945, June). UN Charter. Retrieved April 23, 2024, from <https://www.un.org/en/about-us/un-charter>
- Newman, E., & Visoka, G. (2016). The Foreign Policy of State Recognition: Kosovo’s Diplomatic Strategy to Join International Society: Table 1. *Foreign Policy Analysis*, orw042. <https://doi.org/10.1093/fpa/orw042>
- Oates, S. (2006). *Television, democracy and elections in Russia* [OCLC: 1345642721]. Taylor & Francis.
- Oganisian, A., & Roy, J. A. (2021). A practical introduction to Bayesian estimation of causal effects: Parametric and nonparametric approaches. *Statistics in Medicine*, 40(2), 518–551. <https://doi.org/10.1002/sim.8761>
- OpenAI, Achiam, J., Adler, S., Agarwal, S., Ahmad, L., Akkaya, I., Aleman, F. L., Almeida, D., Altenschmidt, J., Altman, S., Anadkat, S., Avila, R., Babuschkin, I., Balaji, S., Balcom, V., Baltescu, P., Bao, H., Bavarian, M., Belgum, J., ... Zoph, B. (2023). GPT-4 Technical Report [Version Number: 6]. <https://doi.org/10.48550/ARXIV.2303.08774>

- Peisakhin, L., & Rozenas, A. (2018). Electoral Effects of Biased Media: Russian Television in Ukraine. *American Journal of Political Science*, 62(3), 535–550. <https://doi.org/10.1111/ajps.12355>
- Potter, B. (2022). Unrecognized Republic, Recognizable Consequences: Russian Troops in “Frozen” Transnistria. *Journal of Advanced Military Studies*, 2022(special), 168–188. <https://doi.org/10.21140/mcu.2022SIstratcul010>
- Radeljić, B., & Özşahin, M. C. (2023). The inefficiency of EU leverage in Serbia during the Russia-Ukraine war. *Southeast European and Black Sea Studies*, 23(4), 697–716. <https://doi.org/10.1080/14683857.2023.2255029>
- Robinson, P. (2002). *The CNN Effect: The Myth of News, Foreign Policy and Intervention*. Taylor; Francis.
- Rubin, D. B. (1981). The Bayesian Bootstrap. *The Annals of Statistics*, 9(1). <https://doi.org/10.1214/aos/1176345338>
- Slothuus, R., & Bisgaard, M. (2021). How Political Parties Shape Public Opinion in the Real World. *American Journal of Political Science*, 65(4), 896–911. <https://doi.org/10.1111/ajps.12550>
- Snyder, G. H. (1997). *Alliance politics*. Cornell University Press.
- Springman, J., Adiguzel, S., Lin, Z.-R., Moratz, D., Romero, D., Su, H., Villamizar-Chaparo, M., & Wibbels, E. (2026). Foreign Influence by Authoritarian Governments: Introducing New Data and Evidence. *Working Paper*.
- Svolik, M. W. (2012). *The politics of authoritarian rule*. Cambridge University Press.
- Toral, G. (2024). How Patronage Delivers: Political Appointments, Bureaucratic Accountability, and Service Delivery in Brazil. *American Journal of Political Science*, 68(2), 797–815. <https://doi.org/10.1111/ajps.12758>

- Tramontin Shinoki, S., Picchetti, P., & Campos De Xavier Pinto, C. (2024). Difference-in-Discontinuities: Estimation, Inference and Validity Tests. *SSRN Electronic Journal*. <https://doi.org/10.2139/ssrn.4846141>
- USIC. (2024, March). The IC OSINT Strategy, 2024-2026. https://www.dni.gov/files/ODNI/documents/IC_OSINT_Strategy.pdf
- Walker, N. (2023, August). *Conflict in Ukraine: A timeline (2014 - eve of 2022 invasion)* (tech. rep. No. Number CBP 9476). House of Commons Library. <https://researchbriefings.files.parliament.uk/documents/CBP-9476/CBP-9476.pdf>
- Walker, N. (2024, March). *Conflict in Ukraine: A timeline (current conflict, 2022 - present)* (tech. rep.). <https://commonslibrary.parliament.uk/research-briefings/cbp-9847/>
- Weitsman, P. A. (2004). *Dangerous alliances: Proponents of peace, weapons of war*. Stanford University Press.
- Williams, H., & Blum, I. (2018). *Defining Second Generation Open Source Intelligence (OSINT) for the Defense Enterprise*. RAND Corporation. <https://doi.org/10.7249/RR1964>
- Xiong, M., Hu, Z., Lu, X., Li, Y., Fu, J., He, J., & Hooi, B. (2023). Can LLMs Express Their Uncertainty? An Empirical Evaluation of Confidence Elicitation in LLMs [Version Number: 2]. <https://doi.org/10.48550/ARXIV.2306.13063>

Appendix

A. Data and Measurement

The following articles illustrate the three sentiment classifications used in our analysis. Each example shows the translated title and first two sentences, which served as the classifier input. Articles were selected to represent clear cases across different countries and media ownership types.

Positive Sentiment Toward Russia

Serbia	<i>danas.rs</i>	Independent	2022-02-07
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Petković: Russia's help on the issue of Kosovo is invaluable.

“The director of the Government Office for Kosovo and Metohija, Petar Petković, assessed today that for Belgrade, Russia’s support and assistance on the issue of Kosovo is invaluable. Petković said in a conversation with Russian Ambassador to Belgrade Aleksandar Bocan Kharchenko that pressure and repression against the Serbian people [continues].”

Belarus	<i>eng.belta.by</i>	State-owned	2022-04-12
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Lukashenko: Belarus has something to contribute to Russia's might.

“Belarus and Russia will surely endure in the current circumstances. Belarus President Aleksandr Lukashenko expressed confidence about it during negotiations with President of Russia Vladimir Putin at the Vostochny Cosmodrome in Russia’s Amur Oblast on 12 April.”

Neutral Sentiment Toward Russia

Turkey	<i>sozcu.com.tr</i>	Independent	2022-04-19
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Russia and Ukraine traded prisoners.

“Russia and Ukraine made a prisoner exchange. While Russia’s 56th day of occupation was concentrated in the east of the country, the two countries reportedly made a prisoner exchange.”

Armenia	<i>golosarmenii.am</i>	Independent	2022-04-01
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Armenia abstained in the UNHRC vote on unilateral sanctions.

“Twenty-seven of the 47 countries, including Russia, China and India, voted in favour of the relevant resolution adopted at the meeting on Thursday, while 14 States, including the United States, the European Union, the United Kingdom and Ukraine, objected. Six delegations abstained in the vote, including Armenia.”

Negative Sentiment Toward Russia

Turkey	<i>sozcu.com.tr</i>	Independent	2022-03-30
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The Russian-Ukraine war... there are bodies in the streets of Irpin.

“The Russian-Ukraine war... there are bodies on the streets in Irpin, near Kiev. A promising statement from Russia following the talks in Istanbul yesterday, but the images taken at Irpin, which were at the bottom of Kiev, terrified the international public.”

Serbia	<i>danas.rs</i>	Independent	2022-05-26
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De-Ukrainization in Mariupol: Russians installed mobile TV propaganda, children abolished holidays.

“The Russian forces that occupied Mariupol have decided to cancel the summer vacation so that the students can master the Russian program, said today the mayor’s adviser Petro Andryushchenko. Children will have classes in Russian language and literature, Russian history and mathematics all summer.”

The main text reports validation results for the sentiment and topic classifiers. The Russian involvement classifier, which identifies whether Russia is a main actor in an article, achieves an F1 score of 0.83 against the human-coded gold standard. One feature of this classifier merits note: it is more likely than human coders to classify articles as not featuring Russia as a main actor. This measure is used only in supplementary analyses.

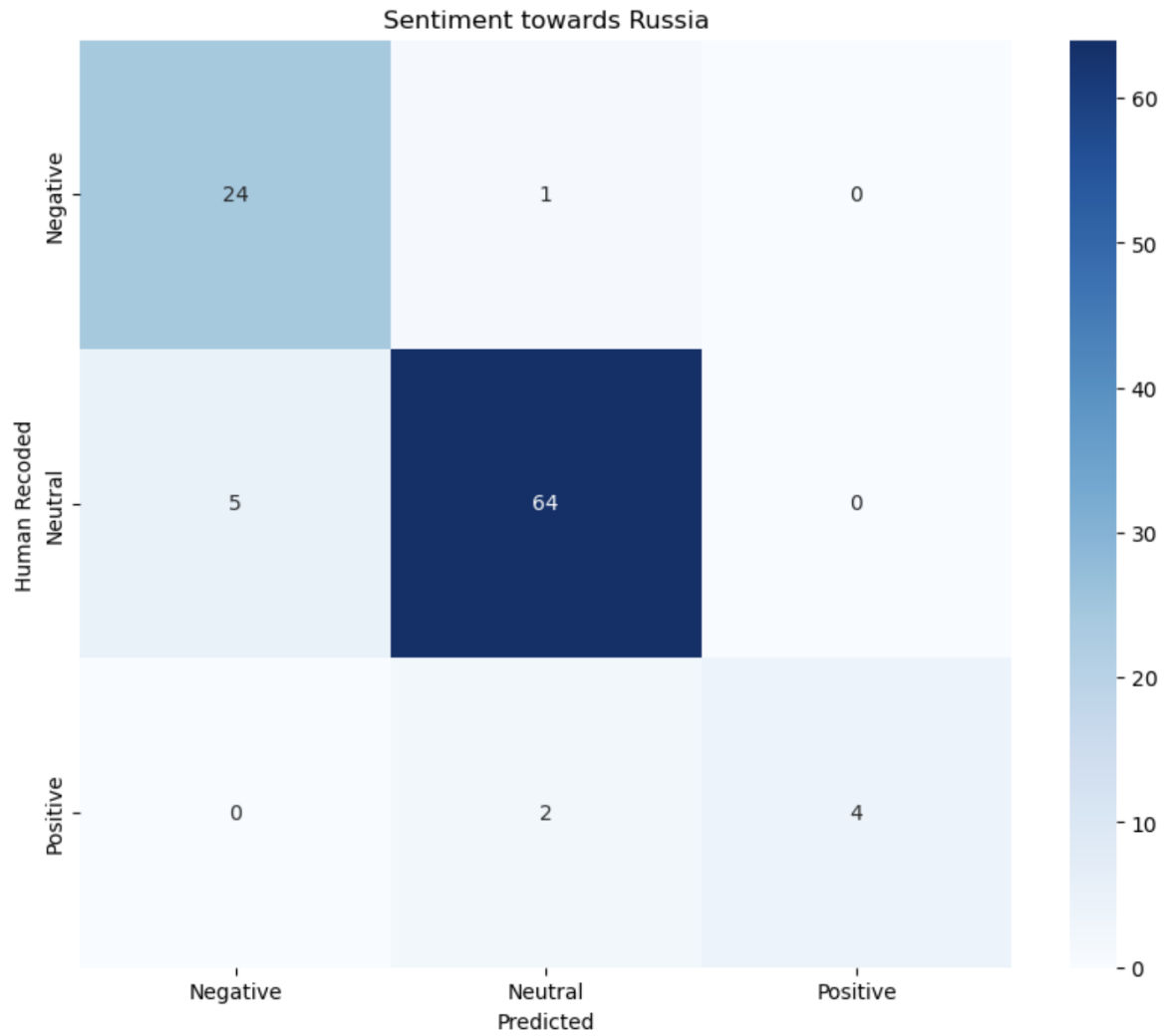


Figure A1: Confusion matrix for the GPT-4o sentiment classifier, evaluated against the human-coded gold standard.

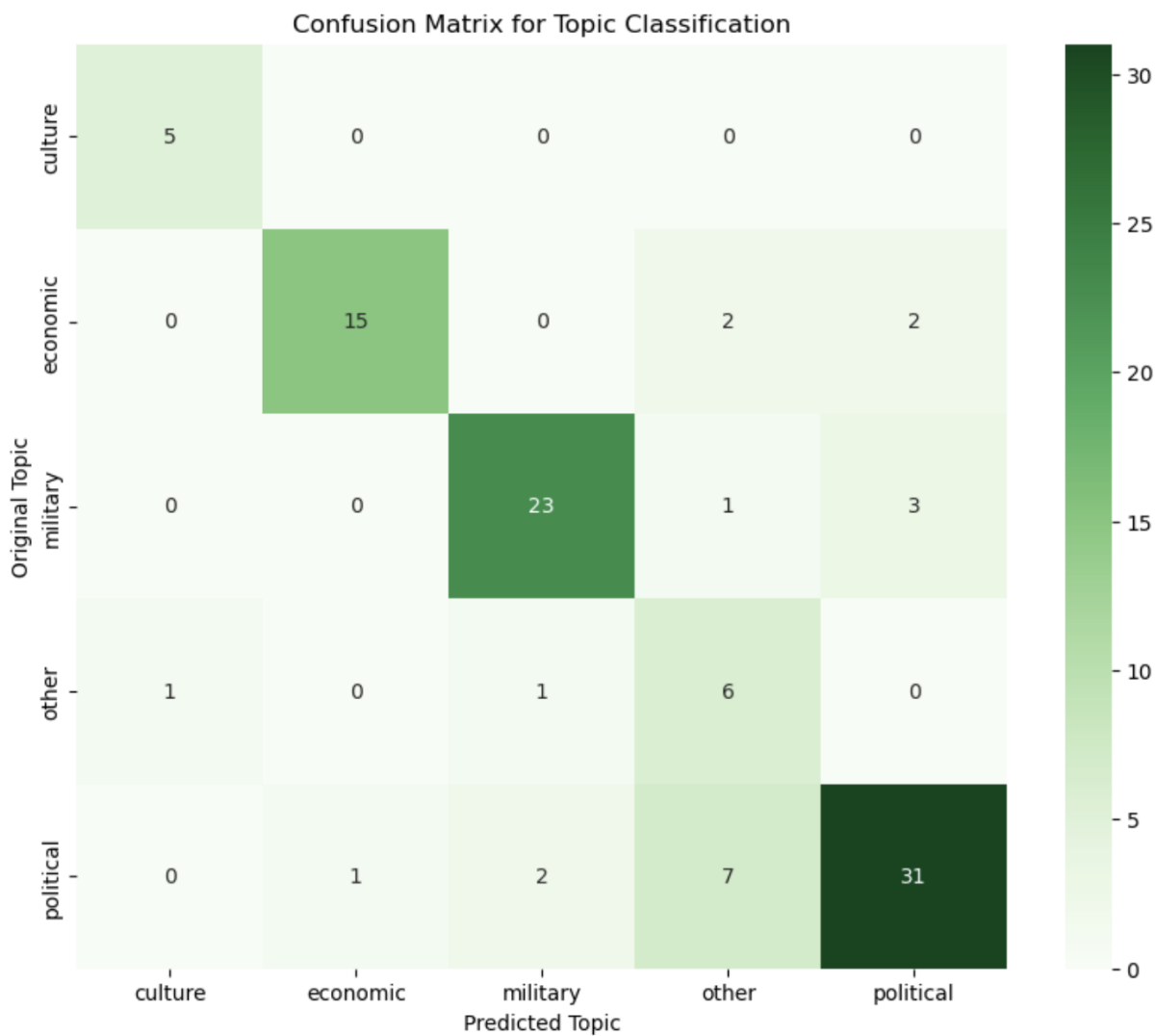


Figure A2: Confusion matrix for the GPT-4o topic classifier, evaluated against the human-coded gold standard.

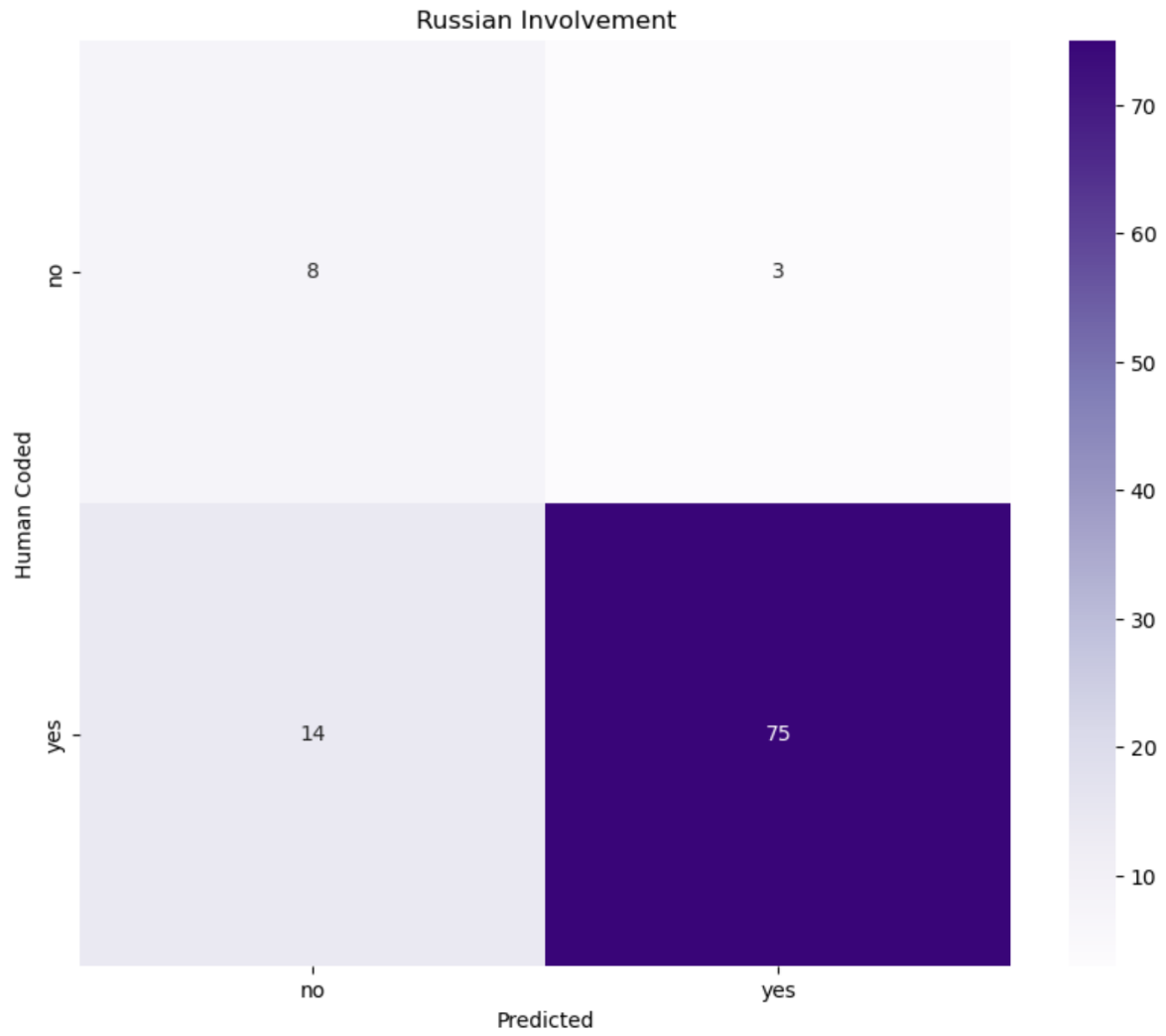


Figure A3: Confusion matrix for the GPT-4o Russian involvement classifier, evaluated against the human-coded gold standard.

B. Additional Descriptive Figures

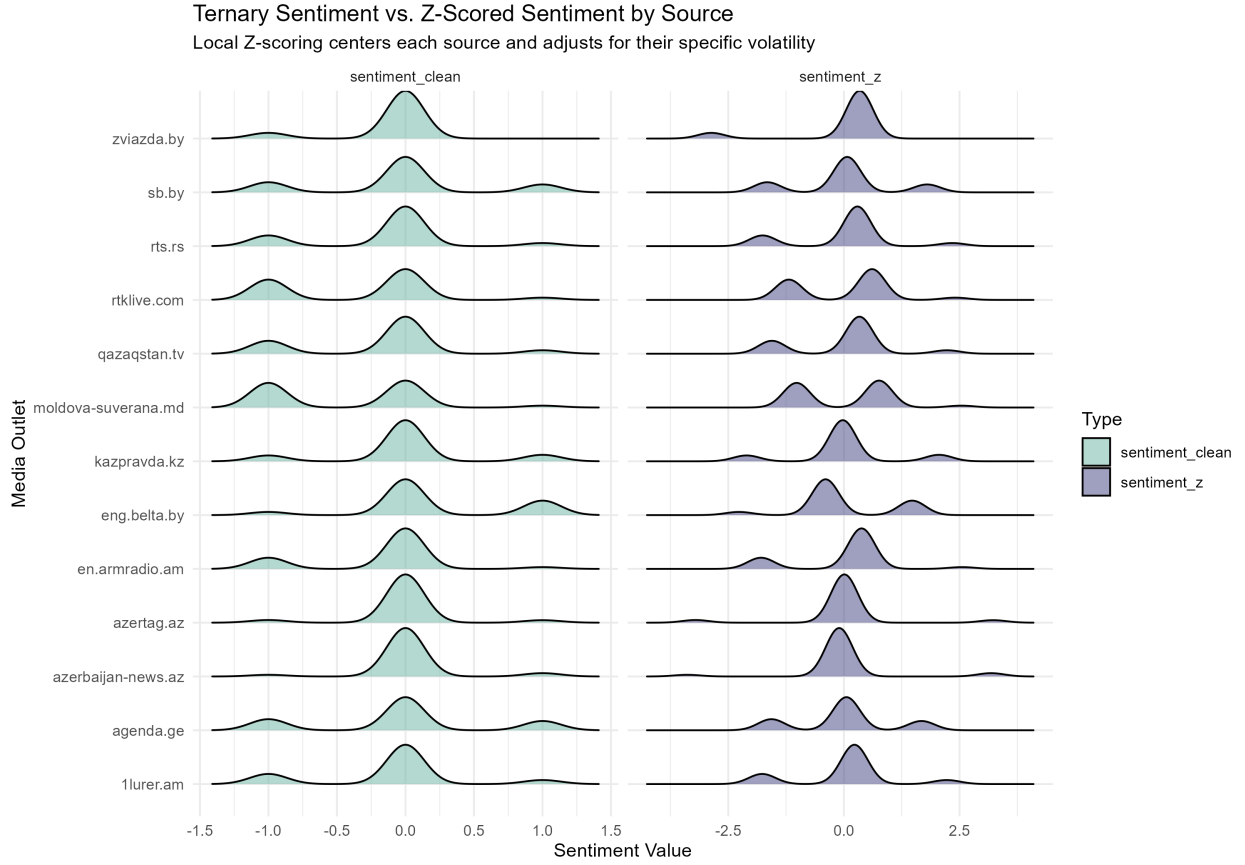


Figure A4: Comparison of raw sentiment and z-scored sentiment distributions across sources, validating the z-score standardization procedure described in the main text.

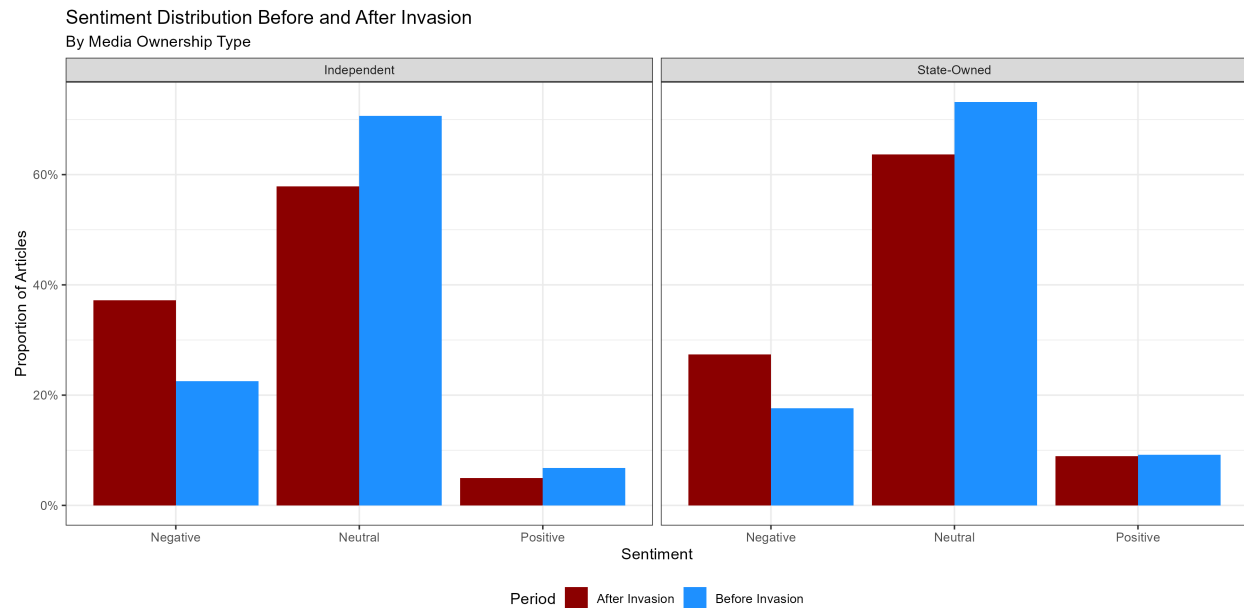


Figure A5: Distribution of sentiment scores across the sample.

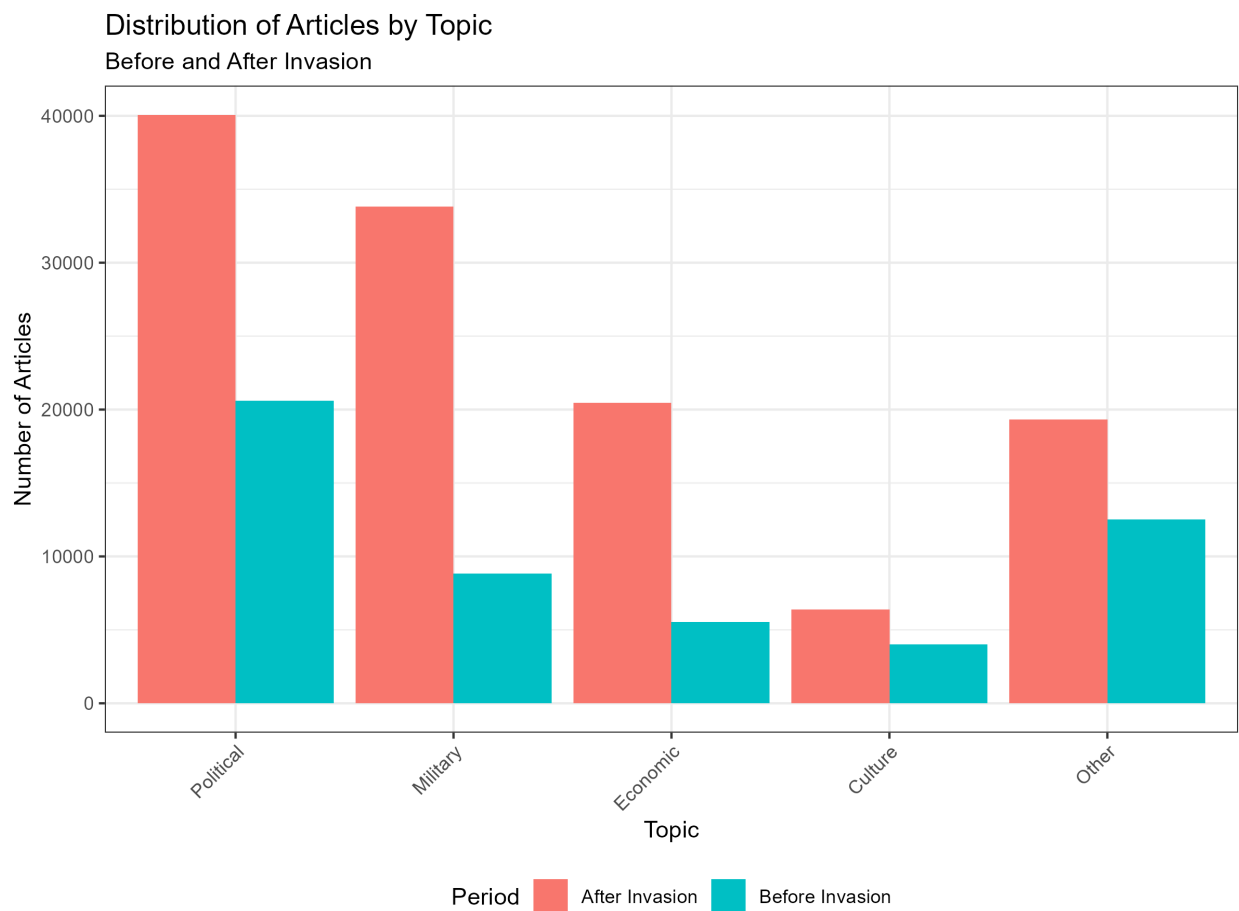


Figure A6: Distribution of articles across topic categories.

C. Extended Research Design

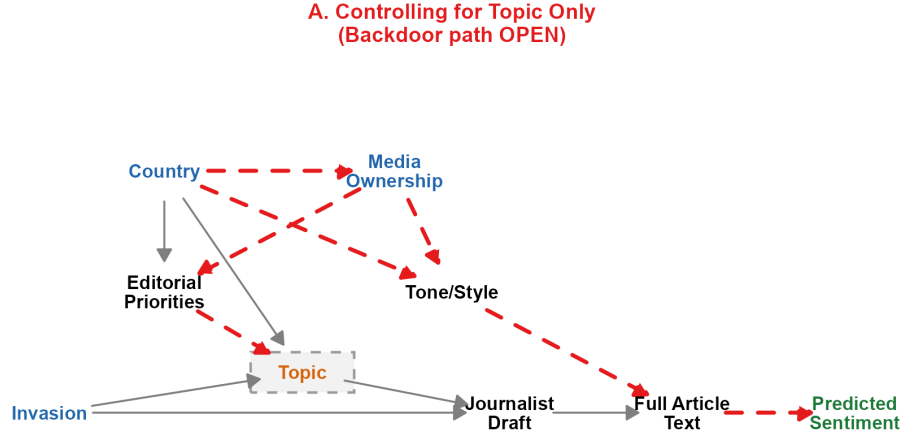


Figure A7: Directed acyclic graph illustrating the identification problem. The invasion affects sentiment both directly and indirectly through changes in topic composition. A naive approach of blocking only on topic introduces a backdoor path through country and ownership.

The topic-adjusted estimates in the main text identify a controlled direct effect. Interpreting this quantity requires an assumption beyond that of continuity, namely sequential unconfoundedness. This assumption carries two components. First is that, conditional on pretreatment confounders, there is no omitted variable that affects both the treatment (here the invasion) and the outcome (sentiment). This component is supplied by the continuity assumption of the RDD. Second is that, conditional on treatment, pretreatment confounders, and intermediate confounders, there is no omitted variable that affects both the mediator (here topic) and the outcome. This assumption is trickier. Day-level war news shifting both topic and sentiment is the leading candidate for a violation here. However, to the extent that these shocks are common, they are differ-

enced out of the interaction in the RDID approach since they would be shared by both groups in our dataset, making this stronger assumption more plausible.

Neutral-Country Media: Starting Level and Drop (Day)

State-owned outlets in neutral countries start close to independent outlets and far above the scale floor, so the absence of a drop is not a floor or measurement artifact.

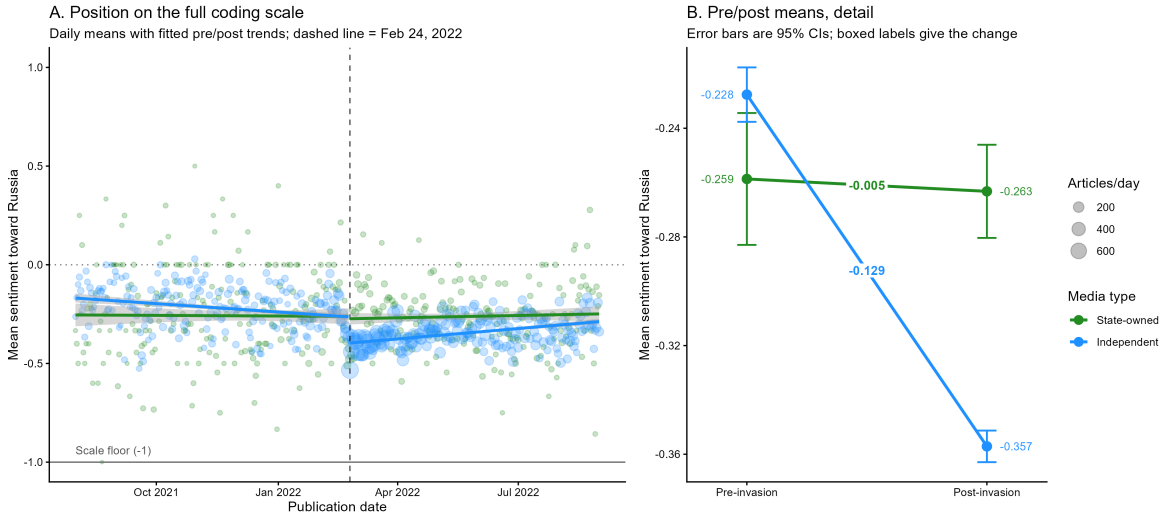


Figure A8: Neutral-country media relative to the coding-scale floor. Panel A places daily mean sentiment for state-owned and independent outlets in neutral countries on the full coding scale; both groups sit far above the scale floor of -1 throughout. Panel B shows pre- and post-invasion period means: state-owned outlets move by -0.005 while independent outlets in the same countries drop by -0.129 . The absence of a state-media response in neutral countries is therefore not a floor or measurement artifact.

Two of the checks summarized in the Robustness section rest on logic worth stating in full. For hypothesis 2, we additionally restrict the sample to non-NATO countries. Because NATO members have a clear institutional basis for opposing the invasion that is distinct from the signaling mechanism we theorize, including them risks conflating alliance-driven opposition with the state-versus-independent media dynamic. For the compositional checks, topic fixed effects may inadequately address the compositional threat if topic composition shifts discontinuously at the cutoff. We therefore omit articles coded as military in nature and re-estimate all specifications: if the main result were driven by the mechanical influx of negatively-valenced military coverage, dropping these articles would attenuate or eliminate the effect. It does not.

Table A1: H1 RDD — Fixed Effects Robustness (Probabilistic Time)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Invasion	-0.159*** (0.011)	-0.1515*** (0.0092)	-0.1398*** (0.0085)	-0.1398*** (0.0085)	-0.243*** (0.016)	-0.091*** (0.015)	-0.147*** (0.013)
Topic FEs	-	-	-	X	X	X	X
State FEs	-	X	X	X	-	X	X
Source FEs	-	-	X	X	-	X	X
Source Z-Score	-	-	-	-	X	-	-
Clusters	62	62	62	62	61	—	—
Num.Obs.	174094	174094	174094	174094	174056	130525	93401
R2	0.024	0.056	0.087	0.087	0.035		
R2 Adj.	0.024	0.056	0.086	0.086	0.035		

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

All standard errors are clustered at the source level. Models 1-4 use the full dataset with progressively added fixed effects. Model 5 uses a Z-Score normalization with the full dataset. Models 6-7 use full fixed effects, restricting to non-military topics and non-NATO countries respectively.

D. Main Results: Additional Specifications

D.1 Hypothesis 1

Table A2: H1 RDD by Topic — Z-Score (Probabilistic Time)

	Political	Military	Economic	Cultural	Other
Invasion	-0.160*** (0.040)	-0.422*** (0.062)	-0.332*** (0.066)	-0.569*** (0.095)	-0.219*** (0.036)
Source Z-Score	X	X	X	X	X
Clusters	—	—	—	—	—
Num.Obs.	61660	43568	26230	10465	32133

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

All standard errors are clustered at the source level. All models use a source-level Z-Score normalization with the optimal bandwidth. Each column restricts to the indicated topic category.

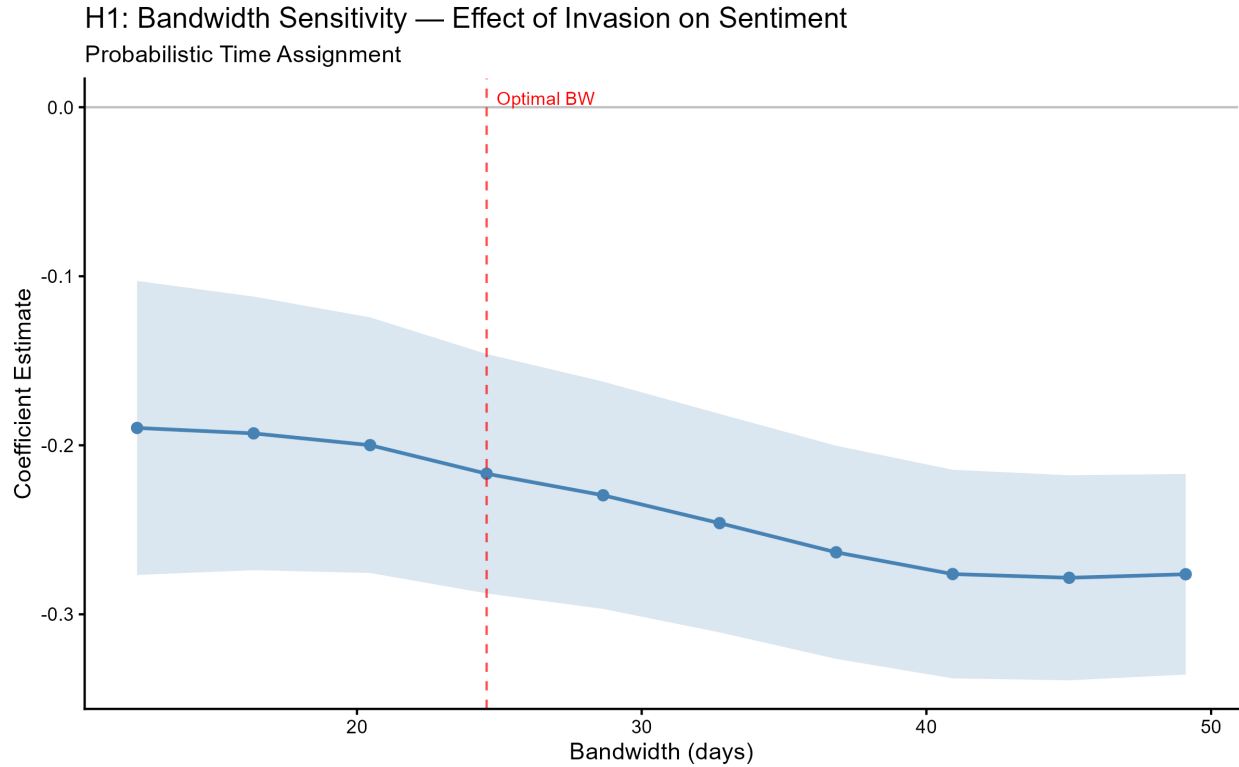


Figure A9: Bandwidth sensitivity for hypothesis 1. Each point reports the RDD coefficient estimate at the indicated bandwidth. The red dashed line marks the MSE-optimal bandwidth. Shaded region shows 95% confidence intervals.

D.2 Hypothesis 2

Table A3: H2 Diff-in-Disc — FE Interaction Core (Probabilistic Time)

	(1)†	(2)†	(3)†	(4)†	(5)†	(6)†	(7)†
Invasion	-0.263*** (0.018)	-0.16*** (0.01)	-0.15*** (0.01)	-0.125*** (0.019)	-0.126*** (0.019)	-0.122*** (0.015)	-0.189*** (0.013)
State Owned x Treatment	0.090** (0.032)	0.066* (0.027)	0.059** (0.018)	0.032 (0.033)	0.046 (0.033)	0.069+ (0.034)	0.146** (0.033)
Source Z-Score	X	-	-	-	-	-	-
State FEs	-	X	X	X	X	X	X
Source FEs	-	-	X	-	X	-	-
Topic FEs	-	X	X	X	X	X	X
Clusters	61	61	61	60	60	60	43
Num.Obs.	174056	174056	174056	37622	37622	27315	32095
R2	0.036	0.061	0.087	0.065	0.084	0.042	0.045
R2 Adj.	0.036	0.061	0.087	0.065	0.082	0.042	0.045

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

All standard errors are clustered at the source level. Model 1 uses a source-level Z-Score normalization with the full dataset. Model 2 uses the full dataset with state + topic FEs. Model 3 adds source FEs. Model 4 uses the optimal bandwidth. Model 5 uses optimal bandwidth with full FEs. Model 6 omits military topics. Model 7 omits NATO allies. Model 8 restricts to Russian-related articles. Model 9 uses matched data. † Standard errors computed via wild cluster restricted (WCR) bootstrap (Cameron, Gelbach & Miller 2008) with Webb six-point weights (B=99999, null imposed), clustered by source domain.

Table A4: H2 Diff-in-Disc — Full FE + DID Robustness (Probabilistic Time)

	(1)†	(2)†	(3)†
Invasion	-0.124*** (0.016)	-0.217*** (0.017)	-0.178*** (0.009)
State Owned x Treatment	0.078+ (0.033)	0.138* (0.038)	0.074** (0.027)
State FEs	X	X	X
Source FEs	X	X	X
Topic FEs	X	X	X
Clusters	60	42	60
Num.Obs.	27315	20604	37622
R2	0.064	0.092	0.057
R2 Adj.	0.062	0.090	0.057

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

All standard errors are clustered at the source level. Model 1 uses non-military subset with full FEs. Model 2 uses Russian-related articles with full FEs. Model 3 is a DID specification within the optimal bandwidth. † Standard errors computed via wild cluster restricted (WCR) bootstrap (Cameron, Gelbach & Miller 2008) with Webb six-point weights (B=99999, null imposed), clustered by source domain.

Table A5: H2 Diff-in-Disc by Topic — Full Data (Probabilistic Time)

	Political	Military	Economic	Cultural	Other
Invasion	-0.175*** (0.027)	-0.316*** (0.029)	-0.31*** (0.06)	-0.44*** (0.05)	-0.216*** (0.042)
State Owned x Treatment	0.076+ (0.041)	0.022 (0.046)	0.086 (0.096)	0.02 (0.13)	0.220** (0.072)
State FEs	X	X	X	X	X
Source FEs	-	-	-	-	-
Topic FEs	-	-	-	-	-
Clusters	61	61	61	61	61
Num.Obs.	61660	43568	26230	10465	32133
R2	0.006	0.019	0.028	0.027	0.009
R2 Adj.	0.006	0.019	0.028	0.026	0.009

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

All standard errors are clustered at the source level. All models use the optimal bandwidth with state fixed effects. Each column restricts to the indicated topic category. † Standard errors computed via wild cluster restricted (WCR) bootstrap (Cameron, Gelbach & Miller 2008) with Webb six-point weights (B=99999, null imposed), clustered by source domain.

Table A6: H2 Leave-One-Out Country Robustness (Probabilistic Time)

Country Dropped	Coefficient	Std. Error	p-value	N	Bandwidth
Standard Model					
Albania	0.171**	(0.060)	0.006	27,709	21.5
Azerbaijan	0.133*	(0.064)	0.044	24,124	18.9
Armenia	0.157*	(0.065)	0.020	23,875	19.4
Belarus	0.152*	(0.066)	0.026	24,849	19.2
Hungary	0.156**	(0.054)	0.005	29,259	24.5
Kazakhstan	0.169**	(0.060)	0.007	30,457	22.8
Kosovo	0.159*	(0.062)	0.013	30,302	27.3
Moldova	0.134*	(0.062)	0.035	29,167	22.0
Serbia	0.121	(0.066)	0.073	25,116	19.7
Turkey	0.210**	(0.074)	0.006	29,326	28.4
Georgia	0.131*	(0.059)	0.031	28,633	21.4
Matched Model					
Albania	0.180**	(0.058)	0.003	27,704	21.5
Azerbaijan	0.147*	(0.064)	0.027	24,121	18.9
Armenia	0.170**	(0.063)	0.009	23,871	19.4
Belarus	0.162*	(0.065)	0.016	24,764	19.2
Hungary	0.165**	(0.054)	0.004	29,248	24.5
Kazakhstan	0.181**	(0.060)	0.004	30,452	22.8
Kosovo	0.170**	(0.064)	0.010	30,292	27.3
Moldova	0.144*	(0.061)	0.022	29,162	22.0
Serbia	0.141*	(0.065)	0.034	25,112	19.7
Turkey	0.216**	(0.075)	0.006	29,322	28.4
Georgia	0.144*	(0.058)	0.017	28,628	21.4

Leave-one-out robustness analysis. Each row shows results when excluding the specified country. Bandwidth is recalculated for each subset. Coefficient shown is for the treatment:state_ownership interaction term. Standard errors clustered at source domain level.

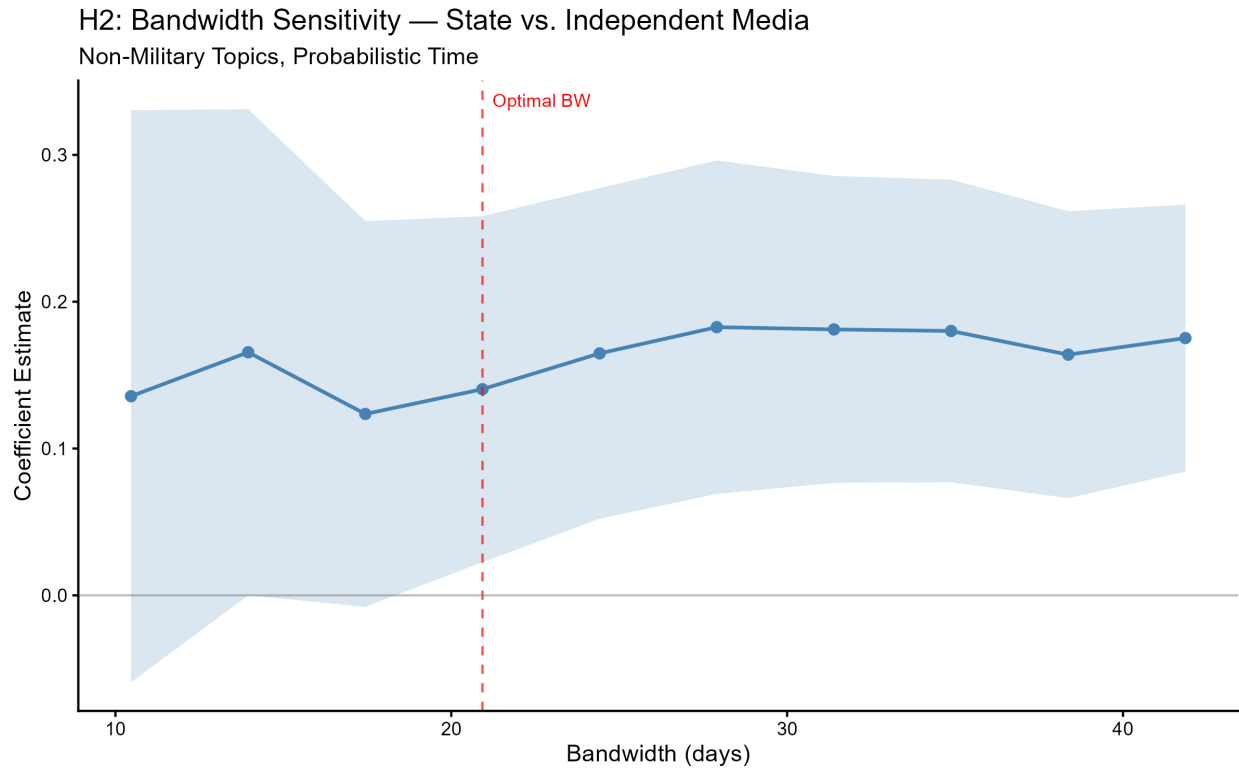


Figure A10: Bandwidth sensitivity for hypothesis 2. Each point reports the RDID interaction coefficient estimate at the indicated bandwidth. The red dashed line marks the MSE-optimal bandwidth. Shaded region shows 95% confidence intervals.

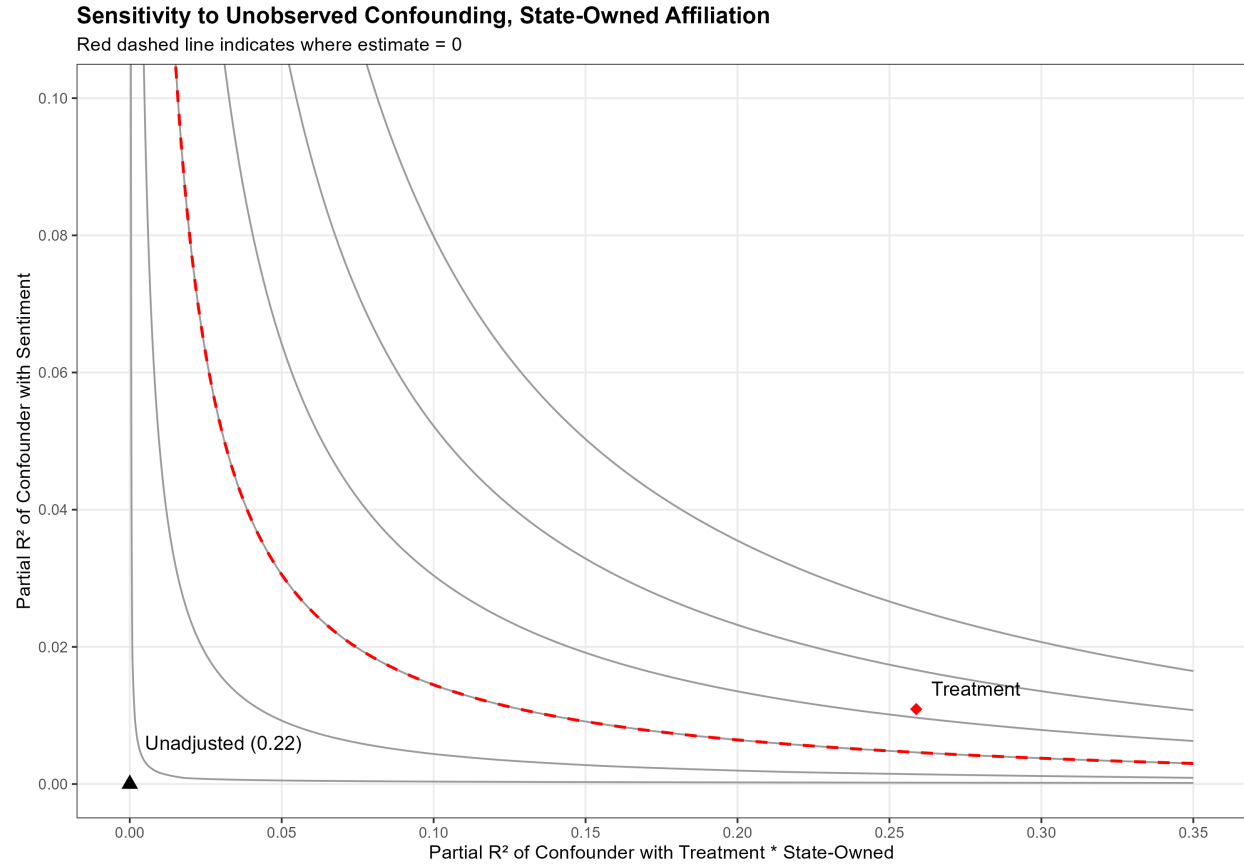


Figure A11: Sensitivity of the state-owned $\times$ treatment interaction to unobserved confounding, following Cinelli and Hazlett (2020). The red dashed line indicates the combinations of confounder strength that would reduce the estimate to zero.

D.3 Hypothesis 3

Table A7: H3a Alignment — FE Interaction + Robustness (Probabilistic Time)

	(1)†	(2)†	(3)†	(4)†	(5)†	(6)†	(7)†	(8)†	(9)†
Invasion	-0.150 (0.094)	-0.081 (0.055)	-0.018 (0.037)	0.079 (0.050)	0.061 (0.051)	-0.042 (0.047)	0.098 (0.069)	0.027 (0.034)	0.029 (0.046)
CSTO x Treatment	-0.021 (0.102)	-0.0025 (0.0579)	-0.107 (0.052)	-0.202* (0.061)	-0.222+ (0.069)	-0.074 (0.058)	-0.38* (0.11)	-0.149+ (0.049)	-0.179 (0.063)
Source Z-Score	X	-	-	-	-	-	-	-	-
State FEs	-	X	X	X	X	X	X	X	X
Source FEs	-	-	-	-	-	X	-	X	X
Topic FEs	-	X	X	X	X	X	X	X	X
Matched	-	-	-	-	-	-	X	-	-
Clusters	13	13	13	13	13	13	13	13	13
Num.Obs.	26911	26911	7008	5917	4651	7008	5721	5917	4651
R2	0.023	0.086	0.059	0.049	0.076	0.094	0.014	0.088	0.120
R2 Adj.	0.023	0.085	0.057	0.047	0.074	0.091	0.013	0.085	0.116

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

All standard errors are clustered at the source level. Sample restricted to state-owned media in non-NATO countries. Model 1 uses a source-level Z-Score normalization with the full dataset. Model 2 uses the full dataset with FEs. Model 3 uses the optimal bandwidth. Model 4 omits military topics. Model 5 restricts to Russian-related articles. Model 6 uses optimal bandwidth with full FEs. Model 7 uses matched data. Models 8-9 add full FEs to non-military and Russian subsets. † Standard errors computed via wild cluster restricted (WCR) bootstrap (Cameron, Gelbach & Miller 2008) with Webb six-point weights (B=99999, null imposed), clustered by source domain.

Table A8: H3a Alignment by Topic — Z-Score (Probabilistic Time)

	Political	Military	Economic	Cultural	Other
Invasion	0.0051 (0.0725)	-0.46+ (0.25)	0.15 (0.11)	-1.35** (0.34)	0.33 (0.60)
CSTO x Treatment	-0.26 (0.12)	0.29 (0.28)	-0.39 (0.21)	0.74 (0.45)	-0.32 (0.60)
Source Z-Score	X	X	X	X	X
Clusters	13	13	13	13	13
Num.Obs.	2661	1783	1343	501	2206
R2	0.004	0.038	0.021	0.099	0.028
R2 Adj.	0.001	0.034	0.016	0.086	0.025

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

All models use a source-level Z-Score normalization with the optimal bandwidth. Sample restricted to state-owned media in non-NATO countries. Each column restricts to the indicated topic category. † Standard errors computed via wild cluster restricted (WCR) bootstrap (Cameron, Gelbach & Miller 2008) with Webb six-point weights (B=99999, null imposed), clustered by source domain. ‡ Cluster-robust SEs from rdrobust.

Table A9: H3a Alignment by Topic — Full Data (Probabilistic Time)

	Political	Military	Economic	Cultural	Other
Invasion	-0.085 (0.068)	-0.33* (0.12)	-0.052 (0.154)	-0.86* (0.33)	0.12 (0.39)
CSTO x Treatment	-0.050 (0.098)	0.11 (0.15)	-0.29 (0.17)	0.60 (0.34)	-0.11 (0.39)
State FEs	X	X	X	X	X
Source FEs	-	-	-	-	-
Topic FEs	-	-	-	-	-
Clusters	13	13	13	13	13
Num.Obs.	9049	4376	4884	2401	6201
R2	0.011	0.021	0.018	0.020	0.018
R2 Adj.	0.011	0.019	0.016	0.017	0.017

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

All standard errors are clustered at the source level. Sample restricted to state-owned media in non-NATO countries. All models use the optimal bandwidth with state fixed effects. Each column restricts to the indicated topic category. † Standard errors computed via wild cluster restricted (WCR) bootstrap (Cameron, Gelbach & Miller 2008) with Webb six-point weights (B=99999, null imposed), clustered by source domain.

Table A10: H3a Leave-One-Out Country Robustness (Probabilistic Time)

Country Dropped	Coefficient	Std. Error	p-value	N	Bandwidth
Standard Model					
Belarus	-0.188	(0.136)	0.201	5,664	75.2
Armenia	-0.258*	(0.102)	0.030	4,884	37.2
Georgia	-0.230*	(0.101)	0.043	5,830	41.3
Azerbaijan	-0.361*	(0.136)	0.024	5,232	39.6
Kazakhstan	-0.249*	(0.085)	0.015	5,286	36.3
Moldova	-0.326*	(0.114)	0.015	5,706	39.0
Kosovo	-0.298	(0.138)	0.054	5,046	37.1
Serbia	-0.387**	(0.112)	0.005	5,715	40.2
Matched Model					
Belarus	-0.203	(0.142)	0.186	5,314	75.2
Armenia	-0.344*	(0.109)	0.010	4,589	37.2
Georgia	-0.280*	(0.107)	0.024	5,374	41.3
Azerbaijan	-0.387*	(0.136)	0.017	4,946	39.6
Kazakhstan	-0.277**	(0.076)	0.004	4,971	36.3
Moldova	-0.383**	(0.110)	0.005	5,288	39.0
Kosovo	-0.348*	(0.130)	0.021	4,505	37.1
Serbia	-0.407**	(0.112)	0.004	5,103	40.2

Leave-one-out robustness analysis. Each row shows results when excluding the specified country. Bandwidth is recalculated for each subset. Coefficient shown is for the treatment:csto interaction term. Standard errors clustered at source domain level.

Table A11: Treatment Effect Decomposition by Source

	Estimate	SE (Pairs)	p (Pairs)	SE (Bayes)	p (Bayes)	N
Panel A: Group-Level Effects						
CSTO	-0.211	(0.071)	0.000***	(0.058)	0.000***	9
Non-CSTO	-0.004	(0.134)	0.973	(0.102)	0.988	4
Difference	-0.206	(0.152)	0.230	(0.117)	0.116	13
Panel B: Source-Level Effects						
1lurer.am (CSTO)	-0.354	—	—	—	—	641
azerbaijan-news.az (CSTO)	-0.147	—	—	—	—	125
azertag.az (CSTO)	-0.057	—	—	—	—	625
en.armradio.am (CSTO)	-0.276	—	—	—	—	260
eng.belta.by (CSTO)	-0.129	—	—	—	—	432
kazpravda.kz (CSTO)	-0.853	—	—	—	—	241
qazaqstan.tv (CSTO)	-0.349	—	—	—	—	57
sb.by (CSTO)	-0.156	—	—	—	—	2031
zviazda.by (CSTO)	0.281	—	—	—	—	41
agenda.ge (Neutral)	0.150	—	—	—	—	375
moldova-suverana.md (Neutral)	-0.221	—	—	—	—	215
rtklive.com (Neutral)	0.144	—	—	—	—	732
rts.rs (Neutral)	-0.387	—	—	—	—	315

Panel A reports N-weighted average treatment effects by CSTO group, estimated from separate RDD regressions within each source-topic cell. Pairs bootstrap resamples sources with replacement ($B = 9,920$). Bayesian bootstrap draws Dirichlet(1,...,1) source weights (Rubin 1981; $B = 9,920$). Panel B reports source-level treatment effects (N-weighted average across topics within each source). N in Panel A is number of sources; N in Panel B is articles per source. Sample restricted to state-owned, non-NATO, non-military articles within the optimal bandwidth. Source-level Z-score normalization applied.

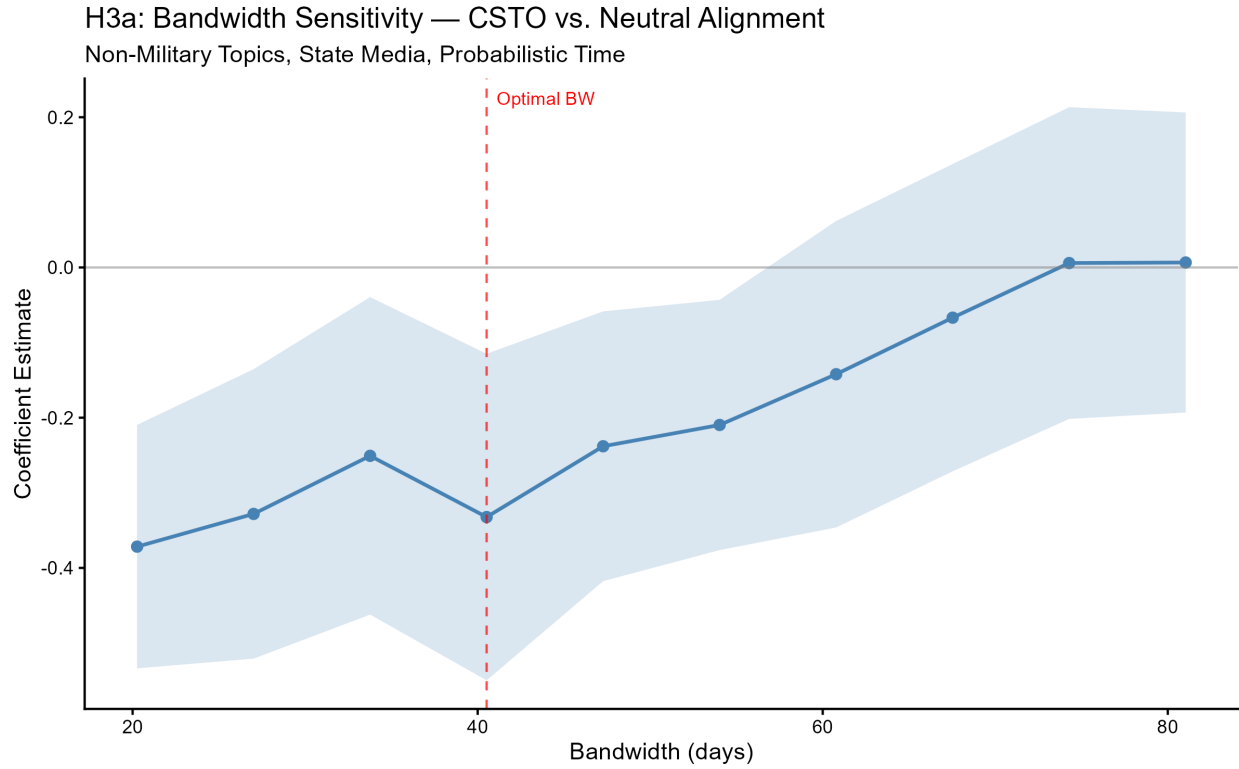


Figure A12: Bandwidth sensitivity for hypothesis 3. Each point reports the RDID interaction coefficient estimate at the indicated bandwidth. The red dashed line marks the MSE-optimal bandwidth. Shaded region shows 95% confidence intervals.

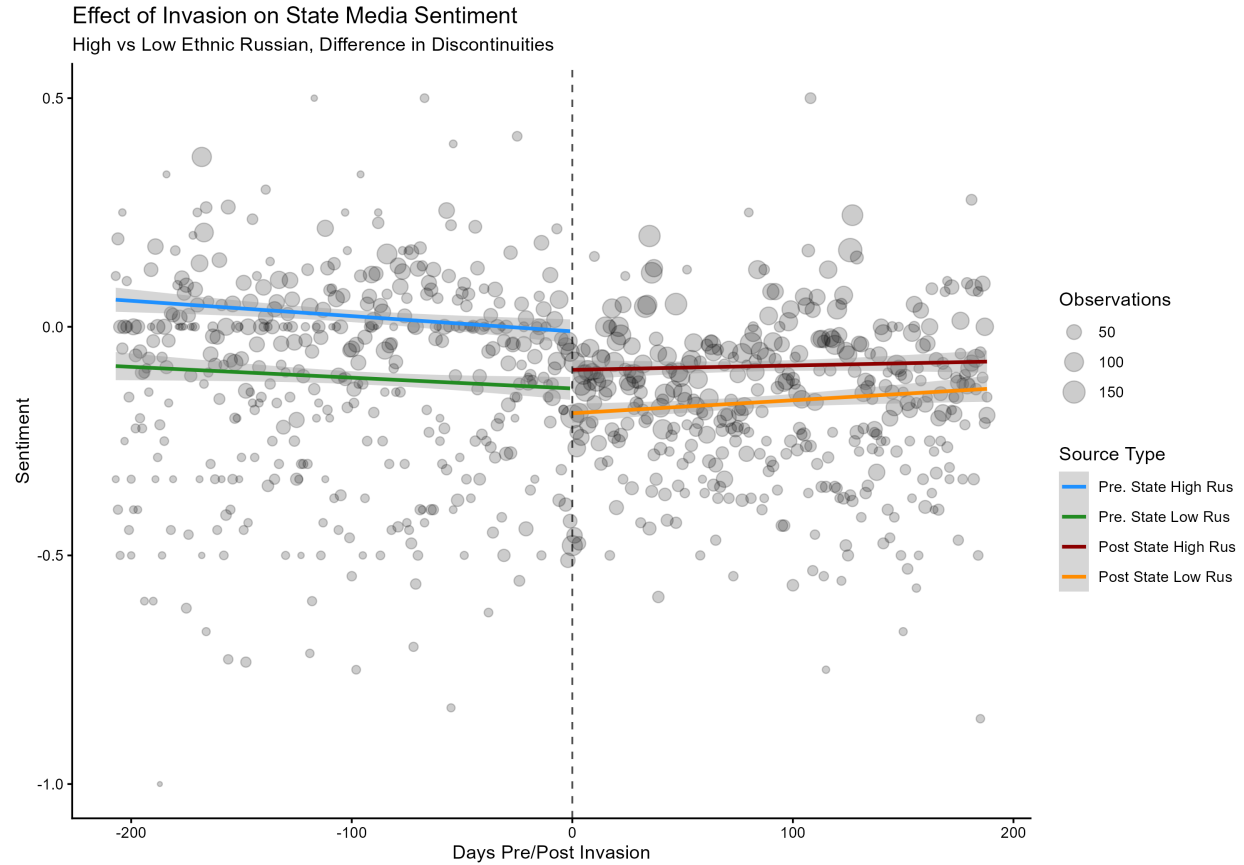


Figure A13: RDD estimates of the effect of the invasion on sentiment toward Russia in state-owned media, by ethnic Russian population share. Sample restricted to state-owned media in non-NATO countries. Each point represents one day, sized by the number of articles published. The dashed line marks February 24th, 2022.

E. Placebo Tests

Table A12: Placebo Test Summary (Probabilistic Time). December 7, 2021 False Treatment Date

Model	Estimate	Std. Error	p-value	Bandwidth	N
(1) H1: Full Model (RDD)	0.016	(0.033)	0.637	771160.0	3,492
(2) H1: Z-Score (RDD)	0.000	(0.063)	0.994	901739.1	4,865
(3) H2: Diff-in-Disc	0.250	(0.119)	0.087	8.9	4,691
(4) H3: Non-Military	0.705	(0.287)	0.143	21.3	1,983

Placebo tests using December 7, 2021 (Biden’s announcement) as a false treatment date. Data restricted to pre-invasion period (before Feb 19, 2022). Model 1 uses rdrobust; remaining models use felm with optimal bandwidth. Non-significant treatment effects suggest parallel trends assumption holds.

Table A13: Placebo Test: Russia-Aligned State Media by Country and Pooled (Probabilistic Time). December 7, 2021 False Treatment Date

	Armenia	Azerbaijan	Belarus	Kazakhstan	All Aligned
Estimate	0.095	-0.163	0.320*	-0.172	0.108
Std. Error	(0.223)	(0.143)	(0.143)	(0.265)	(0.089)
95% CI	[-0.343, 0.533]	[-0.444, 0.117]	[0.040, 0.599]	[-0.692, 0.349]	[-0.066, 0.282]
p-value	0.669	0.254	0.025	0.518	0.225
Bandwidth	21.8	18.7	20.9	19.5	27.9
N	325	325	825	128	2,061

Placebo tests using December 7, 2021 (US public warning of the buildup) as a false treatment date, restricted to state-owned media in the Russia-aligned countries, non-military topics, pre-invasion period (before Feb 19, 2022). Each column is a separate rdrobust fit on the source-level z-scored outcome with the MSE-optimal bandwidth. Columns 1-4 are estimated country by country; the final column pools all four aligned countries. Standard errors are rdrobust default (nearest-neighbor heteroskedasticity-robust); source clustering is not used, because each country has only 2-3 state-owned sources and the pooled column is reported on the same basis for comparability. Non-significant discontinuities indicate no pre-invasion jump at the warning date.

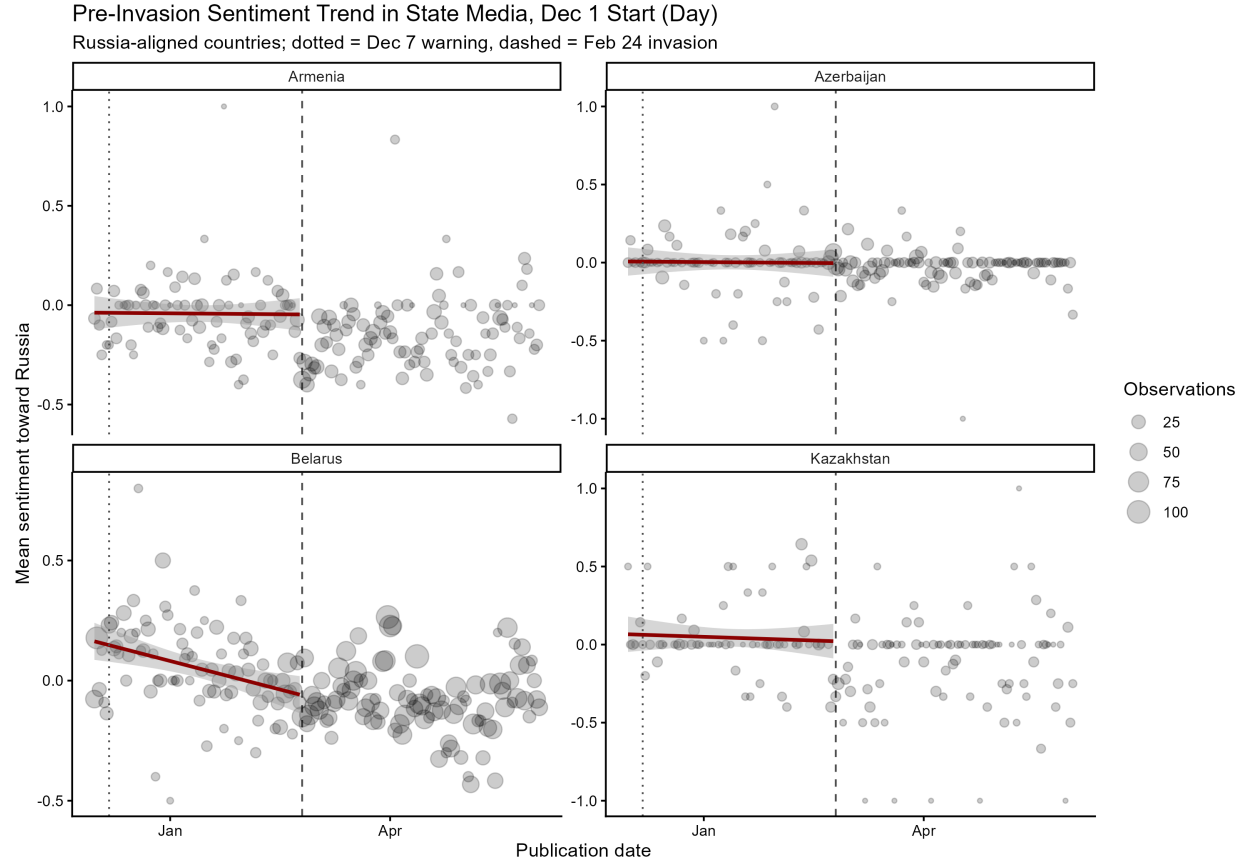


Figure A14: Pre-invasion sentiment in Russia-aligned state media beginning December 1st, 2021. The dotted line marks the December 7th intelligence warning and the dashed line marks the February 24th, 2022 invasion. Three of these trends are indistinguishable from zero, while Belarus declines from its elevated post-announcement level in this window. This is consistent with the interpretation discussed in the main text.

F. Total Effects and Pre-Period Adjustments

Tables in this section report the total-effect comparisons and pre-period adjustment variants discussed in the Robustness section.

Table A14: Total Effect Without Topic Adjustment — H1 (Probabilistic Time)

	(1) Total‡	(2) Main‡
Invasion	-0.240*** (0.035)	-0.217*** (0.036)
Topic FEs	-	X
Source Z-Score	X	X
N (within bandwidth)	45,398	45,366
Clusters	—	—

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

The total-effect model removes the topic adjustment. It is a distinct estimand, not a robustness check: it lets the topic composition of coverage move rather than holding it fixed. Outcome, source clustering and bandwidth procedure are otherwise unchanged, with the bandwidth re-selected without topic. Model 2 is Table 1 Model 5, the main H1 specification, which already uses all articles including military coverage, so both models share a sample and no separate comparator is needed. ‡ Cluster-robust SEs from rdrobust.

Table A15: Total Effect Without Topic Adjustment — H2 (Probabilistic Time)

	(1) Non-NATO Sample†	(2) Total†	(3) Main†	(4) Total (Non-Mil.)†
Invasion	-0.355*** (0.039)	-0.382*** (0.042)	-0.347*** (0.027)	-0.339*** (0.028)
State-Owned x Treat	0.17 (0.08)	0.174 (0.088)	0.222* (0.069)	0.208* (0.071)
R2	0.039	0.020	0.024	0.015
R2 Adj.	0.039	0.020	0.023	0.015
Topic FEs	X	-	X	-
Military Included	X	X	-	-
Source Z-Score	X	X	X	X
N (within bandwidth)	38,264	38,218	31,405	31,251
Clusters	43	43	43	43

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

The total-effect model removes the topic adjustment. It is a distinct estimand, not a robustness check: it lets the topic composition of coverage move rather than holding it fixed. Outcome, source clustering and bandwidth procedure are otherwise unchanged, with the bandwidth re-selected without topic. Model 1, the non-NATO sample model, is unreported in the main text. It is the hypothesis 2 interaction estimated on the non-NATO sample, but with all topics included. Model 3 is Table 2 Model 5, the main H2 specification. Models 2 and 4 are unreported in the main text. The four models form a 2x2 of sample against topic adjustment. Model 1 against Model 2, and Model 3 against Model 4, isolate the topic adjustment; Model 1 against Model 3, and Model 2 against Model 4, isolate the exclusion of military coverage. Model 2 is the total effect on all topics, which is the estimand of interest, since the post-invasion shift in topic composition is largely toward military coverage; Model 4 gives the same estimand on the main specification's own sample. † Standard errors computed via wild cluster restricted (WCR) bootstrap (Cameron, Gelbach & Miller 2008) with Webb six-point weights (B=99999, null imposed), clustered by source domain.

Table A16: Total Effect Without Topic Adjustment — H3a (Probabilistic Time)

	(1) Non-NATO Sample†	(2) Total†	(3) Main†	(4) Total (Non-Mil.)†
Invasion	-0.068 (0.080)	-0.053 (0.076)	0.098 (0.070)	0.11 (0.08)
Russia-Aligned x Treat	-0.19 (0.11)	-0.22 (0.12)	-0.33* (0.11)	-0.37* (0.12)
R2	0.025	0.010	0.015	0.008
R2 Adj.	0.024	0.009	0.013	0.007
Topic FEs	X	-	X	-
Military Included	X	X	-	-
Source Z-Score	X	X	X	X
N (within bandwidth)	7,164	6,999	6,094	6,028
Clusters	13	13	13	13

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

The total-effect model removes the topic adjustment. It is a distinct estimand, not a robustness check: it lets the topic composition of coverage move rather than holding it fixed. Outcome, source clustering and bandwidth procedure are otherwise unchanged, with the bandwidth re-selected without topic. Sample is state-owned media in non-NATO countries. Model 1 is Table 4 Model 3; Model 3 is Table 4 Model 4, the main H3a specification. Models 2 and 4 are unreported in the main text. The four models form a 2x2 of sample against topic adjustment. Model 1 against Model 2, and Model 3 against Model 4, isolate the topic adjustment; Model 1 against Model 3, and Model 2 against Model 4, isolate the exclusion of military coverage. Model 2 is the total effect on all topics, which is the estimand of interest, since the post-invasion shift in topic composition is largely toward military coverage; Model 4 gives the same estimand on the main specification's own sample. † Standard errors computed via wild cluster restricted (WCR) bootstrap (Cameron, Gelbach & Miller 2008) with Webb six-point weights (B=99999, null imposed), clustered by source domain.

Table A17: Pre-Period Adjustment Variants — H2, State Ownership, Matched (Probabilistic Time)

	(1) Main†	(2) Topic†	(3) Z-Score†	(4) Both†
Invasion	-0.348*** (0.026)	-0.356*** (0.026)	-0.39*** (0.03)	-0.399*** (0.029)
State-Owned x Treat	0.222* (0.069)	0.237* (0.069)	0.265** (0.064)	0.281** (0.064)
R2	0.024	0.015	0.026	0.018
R2 Adj.	0.023	0.015	0.026	0.018
Topic FEs	X	-	X	-
Pre-Period Topic Adj.	-	X	-	X
Pre-Period Z-Score	-	-	X	X
N (matched)	31,379	31,379	31,379	31,379
Clusters	43	43	43	43

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Model 1 is the published specification. Model 2 removes the topic fixed effects and demeans the outcome by its pre-invasion topic means. Model 3 re-standardizes the outcome on pre-invasion source means and standard deviations, keeping topic fixed effects. Model 4 applies both. Sample is the exact-matched non-NATO, non-military sample of Table 2 Model 6. The interaction is state-owned x invasion. All four use the identical exact-matched sample, bandwidth and source clustering. † Standard errors computed via wild cluster restricted (WCR) bootstrap (Cameron, Gelbach & Miller 2008) with Webb six-point weights (B=99999, null imposed), clustered by source domain.

Table A18: Pre-Period Adjustment Variants — H3a, Russia-Aligned, Matched (Probabilistic Time)

	(1) Main†	(2) Topic†	(3) Z-Score†	(4) Both†
Invasion	0.098 (0.069)	0.107 (0.071)	0.086 (0.055)	0.093 (0.055)
Russia-Aligned x Treat	-0.38* (0.11)	-0.38* (0.11)	-0.352** (0.086)	-0.353** (0.084)
R2	0.014	0.007	0.013	0.008
R2 Adj.	0.013	0.006	0.012	0.007
Topic FEs	X	-	X	-
Pre-Period Topic Adj.	-	X	-	X
Pre-Period Z-Score	-	-	X	X
N (matched)	5,721	5,721	5,721	5,721
Clusters	13	13	13	13

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Model 1 is the published specification. Model 2 removes the topic fixed effects and demeans the outcome by its pre-invasion topic means. Model 3 re-standardizes the outcome on pre-invasion source means and standard deviations, keeping topic fixed effects. Model 4 applies both. Sample is the exact-matched state-owned, non-NATO, non-military sample of Table 4 Model 5. The interaction is Russia-aligned x invasion. All four use the identical exact-matched sample, bandwidth and source clustering. † Standard errors computed via wild cluster restricted (WCR) bootstrap (Cameron, Gelbach & Miller 2008) with Webb six-point weights (B=99999, null imposed), clustered by source domain.

G. Alternative Forcing Variable Constructions

Our main specifications use the date-time of publication as the forcing variable, with missing timestamps imputed probabilistically from observed publication-timing distributions. We replicate the main results under two alternative constructions: using publication date rather than date-time as the running variable, and dropping all articles with imputed timestamps. Across constructions, every estimate retains the sign of the main result; hypothesis 1 remains statistically significant under all three, while hypothesis 2 falls short under the drop construction (bootstrap $p = 0.10$) and hypothesis 3a is marginal under the day construction ($p = 0.06$ – 0.10). Differences in sample size and precision reflect the constructions themselves: day-level time introduces probability mass points that shift the selected bandwidth, and dropping articles without recorded timestamps removes roughly a quarter of the sample (Tables A19–A21).

Table A19: Typology Robustness — H1 — Effect of the invasion on sentiment. Same specification under each time-assignment variant

	Probabilistic	Day	Drop
Main (Table 1, Model 5)			
Estimate	-0.217***	-0.267***	-0.156***
Std. Error	(0.036)	(0.034)	(0.041)
Bootstrap p	—	—	—
N	45,366	68,973	31,603
Clusters	—	—	—

Each column re-estimates the same specification under a different time-assignment typology: probabilistic imputation of missing publication times (the primary analysis), the calendar day as the running variable, or articles without a recorded time dropped. Sample sizes differ across columns by construction, since the typologies retain different articles and select their own MSE-optimal bandwidths. The coefficient reported is Invasion. Table 1 Model 5 is an rdrobust fit reported with cluster-robust standard errors, so no bootstrap p-value applies. Stars follow the bootstrap p-value where one exists.

Table A20: Typology Robustness — H2 — State-owned vs independent media. Same specification under each time-assignment variant

	Probabilistic	Day	Drop
Main (Table 2, Model 5)			
Estimate	0.222*	0.248**	0.154
Std. Error	(0.069)	(0.056)	(0.076)
Bootstrap p	0.032	0.008	0.100
N	31,405	36,502	23,176
Clusters	43	43	32

Matched (Table 2, Model 6)			
Estimate	0.222*	0.248**	0.153
Std. Error	(0.069)	(0.056)	(0.076)
Bootstrap p	0.031	0.008	0.103
N	31,379	36,472	23,164
Clusters	43	43	32

Each column re-estimates the same specification under a different time-assignment typology: probabilistic imputation of missing publication times (the primary analysis), the calendar day as the running variable, or articles without a recorded time dropped. Sample sizes differ across columns by construction, since the typologies retain different articles and select their own MSE-optimal bandwidths. The coefficient reported is State-Owned x Treat. † Standard errors computed via wild cluster restricted (WCR) bootstrap (Cameron, Gelbach & Miller 2008) with Webb six-point weights (B=99999, null imposed), clustered by source domain. Stars follow the bootstrap p-value where one exists.

Table A21: Typology Robustness — H3a — Russia-aligned vs neutral state media. Same specification under each time-assignment variant

	Probabilistic	Day	Drop
Main (Table 4, Model 4)			
Estimate	-0.332*	-0.265+	-0.364*
Std. Error	(0.111)	(0.101)	(0.109)
Bootstrap p	0.049	0.095	0.046
N	6,094	6,469	5,812
Clusters	13	13	11
Matched (Table 4, Model 5)			
Estimate	-0.380*	-0.311+	-0.364*
Std. Error	(0.112)	(0.104)	(0.109)
Bootstrap p	0.027	0.058	0.047
N	5,721	6,097	5,812
Clusters	13	13	11

Each column re-estimates the same specification under a different time-assignment typology: probabilistic imputation of missing publication times (the primary analysis), the calendar day as the running variable, or articles without a recorded time dropped. Sample sizes differ across columns by construction, since the typologies retain different articles and select their own MSE-optimal bandwidths. The coefficient reported is Russia-Aligned x Treat. † Standard errors computed via wild cluster restricted (WCR) bootstrap (Cameron, Gelbach & Miller 2008) with Webb six-point weights (B=99999, null imposed), clustered by source domain. Stars follow the bootstrap p-value where one exists.